

Supplementary Materials:

**TITLE:** Does Supplying a Directory of Qualified Racial Minority Candidates Boost Organizational Diversity? A Field Experiment

**Data and Code Availability:** Field-study data and analysis code, aggregate online-study data and analysis code, and extended tables are available in our replication repository at <https://anonymous.4open.science/r/directory-diversity-replication/>.

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## Section 1: Intent-to-Treat Robustness Test of Primary Analyses

We conducted a pre-registered intent-to treat (ITT) robustness test of our main analysis using the full randomized sample of 1,881 seminars (not just the 1,686 seminars that continued to hold events in the 2024-25 academic year and to list speakers online, allowing us to measure our pre-registered outcomes of interest). In these analyses, we assumed zero Black speakers appeared in missing seminars. Table S6 reports the regression-estimated effects of assignment to the URM Directory Treatment condition on Black speaker outcomes among the full randomized sample ( $N=1,881$ ) using our primary pre-registered regression specifications with both extended controls (Models 2, 4, and 6) and with more limited controls (Models 1, 3, and 5). We find that assignment to treatment significantly increased the percentage of Black speakers per seminar (Table S6, Model 2,  $B = 0.592$ ,  $SE = 0.263$ ,  $p = .025$ ), the number of Black speakers per seminar (Table S6, Model 4,  $B = 0.066$ ,  $SE = 0.033$ ,  $p = .049$ ), and the likelihood that a seminar featured at least one Black speaker (Table S6, Model 6,  $B = 0.044$ ,  $SE = 0.020$ ,  $p = .030$ ) in a seminar series. These results suggest that our main findings are robust to the inclusion of unobserved seminars under conservative assumptions about their outcomes.

We conducted analogous ITT analyses for Underrepresented Minority (URM) and Hispanic speaker outcomes. Consistent with the lack of robust main effects reported in the main text, treatment effects on URM and Hispanic speaker representation remained non-significant under the ITT specification across all outcome measures (all  $p$ 's  $> .13$ ; see Extended Tables E15–E16 in the replication repository).

## Section 2: Demographic Classification

### 2.1 Classification of Seminar Speakers

We classified the race and gender of seminar speakers using a two-pronged approach that combined facial image analysis with name-based inference. For each of the 17,712 unique speakers in our sample (appearing across 23,455 seminar-speaker slots), we first attempted to retrieve a photograph via automated Google Image searches using the speaker's name and institutional affiliation as search terms. Retrieved photographs were then analyzed using DeepFace (Parkhi et al., 2015), an open-source facial analysis framework that estimates demographic attributes—including race and gender—from facial features. In parallel, we generated name-based demographic predictions using OpenAI's GPT-4o model, which inferred each speaker's likely race and gender from their full name.

The GPT-4o model was queried at temperature 0.1 and prompted to respond in JSON format, providing a race and gender classification along with a confidence score for each. The exact prompt supplied to the model was as follows.

System message: You are an expert in demographic analysis based solely on names. You MUST always provide a definitive answer choosing from the allowed categories only. Never use 'unknown' or refuse to make a determination.

User message:

Analyze the demographic characteristics based solely on this name:

Name: {name}

Please provide your best assessment of:

1. Gender - Use ONLY these categories:

- Male
- Female

(You MUST choose one - no unknown/non-binary options)

2. Race/Ethnicity - Use ONLY these categories:

- white (includes European and Middle Eastern origins)
- black (African or African American origins)
- latino (Hispanic or Latino origins)
- asian (includes East Asian, South Asian, and Indian origins)
- native american (Native American origins)

(You MUST choose one - no unknown/mixed/other options)

IMPORTANT:

- You MUST make a determination even if uncertain - choose the most likely option
- If the name could belong to multiple categories, choose the single most likely one
- Use lower confidence scores (0.3-0.5) when uncertain, but still make a choice

Respond in JSON format:

```
{
  "gender": "your_assessment",
  "gender_confidence": 0.0-1.0,
  "race": "your_assessment",
  "race_confidence": 0.0-1.0,
  "reasoning": "brief explanation"
}
```

Both the face-based and name-based approaches produced categorical predictions for race and gender (male or female). The face-based model (DeepFace) natively outputs six race categories (White, Black, Asian, Indian, Middle Eastern, and Latino Hispanic). Our classification pipeline collapses Indian into Asian and Middle Eastern into White, so four DeepFace-derived categories enter the combined classification, with Native American contributed only by the name-based (GPT-4o) path. The name-based model (GPT-4o) classified race into five categories (White, Black, Hispanic/Latino, Asian, or Native American). Demographic labels were assigned using a hierarchical decision rule that prioritized face-based predictions when the facial analysis produced a confidence score exceeding 90% (this was the case for 38% of speakers in our sample); otherwise, we relied on the name-based prediction algorithm (this was the case for 62% of speakers in our sample). This approach reflects prior work showing that name-based race inference is inherently probabilistic and varies substantially in accuracy across racial groups (Imai & Khanna, 2016; Kozlowski et al., 2022), motivating the prioritization of face-based predictions when a high-confidence facial image is available, while ensuring broad coverage through name-based inference for remaining cases.

To confirm the validity of this algorithmic classification, a research assistant independently coded the race and gender of a random subset of 200 speakers and achieved 100% agreement with the automated classifications. Throughout the paper and this appendix, we use the term “Black” to refer to individuals classified as “Black non-Hispanic.”

## **2.2 Classification of Seminar Organizers and Department Chairs**

We classified the race and gender of 567 department chairs and 1,019 seminar organizers using the same two-pronged approach described in Section 2.1. Where classification relied on the name-based path, recipients were classified with an abbreviated variant of the GPT-4o prompt described in Section 2.1, using the same race and gender category lists but without the reasoning field. The full text of this abbreviated prompt is available in the replication package. Face-based predictions met the 90% confidence threshold for 32% of department chairs and 35% of seminar organizers; the remainder were classified via name-based inference. Of the 1,019 seminar organizers, 49 (4.8%) could not have their race and gender classified because the seminar contact listed was a generic departmental email address (e.g., physics-seminar@university.edu) rather than a named, individual academic; these observations were excluded when computing department-level recipient demographic covariates. All 567 department chairs’ demographics were successfully classified.

## **2.3 Classification of Faculty Listed in the URM Directory**

The 550 faculty members in the URM directories provided in our treatment condition had their demographics classified manually by research assistants as part of the directory construction process described in Appendix Section 3.1 (below). Each faculty member's race (Black, Hispanic, or Native American) and gender were recorded by research assistants based on information available on each faculty member's departmental profile, including photographs and

biographical details. To validate these classifications, an independent research assistant who was not involved in the original directory construction coded the race and gender of a random sample of 100 faculty members who were included in the original URM Directory. The research assistant was shown each faculty member's photograph, name, title, university and website, and encouraged to search for any additional information that could help them verify the faculty member's demographic identity. Agreement between the independent RA's coding and the original directory classifications was 92% for race and 98% for gender.

### Section 3: Directory Construction

As described in the main manuscript, seminar organizers and department chairs from each department in the treatment and control conditions were provided with access to a discipline-specific directory of underrepresented minority (URM) faculty or a list of peer department homepages, respectively. The directories were constructed as follows.

#### 3.1 URM Directory (Treatment Condition)

Departments assigned to the *URM Directory Treatment* condition received access to a curated, discipline-specific directory of URM faculty at peer institutions (see Figure S3 for an example). Peer departments were defined as those ranked within plus or minus 20 positions of the host department in the 2024 U.S. News & World Report discipline-specific rankings, with the peer set iteratively expanded one upward and then downward until at least 20 URM faculty at peer departments were identified for inclusion. A minimum of 40 peer departments was ensured by expanding the set of universities near the top or bottom of the ranking, so the count of peer departments used as a control variable occasionally exceeds 40.

Five discipline-specific lists of URM faculty were constructed from which the set of 20+ peer faculty were drawn for each personalized directory shared with a given department chair and seminar organizer. One list was created for each STEM field in the sample: Chemistry (a list of 108 URM faculty), Computer Science (a list of 94 URM faculty), Mathematics (a list of 105 URM faculty), Mechanical Engineering (a list of 131 URM faculty), and Physics (a list of 112 URM faculty), yielding 550 URM faculty entries in total across all lists. Of all the faculty included on these lists, 268 (48.7%) were classified as Black, 280 (50.9%) as Hispanic, and 2 (0.4%) as Native American; 360 (65.5%) were men and 190 (34.5%) were women. The average

directory listing URM faculty from peer departments that we sent to a seminar organizer and department chair included 31 URM faculty (min = 20, max = 64).

To construct the lists of URM faculty that populated the personalized directories shared with those in our treatment condition, three research assistants systematically visited the department websites of all 146 R1 universities in our sample and identified faculty who were underrepresented racial minorities—defined as Black, Hispanic, or Native American—by reviewing faculty profile pages and photographs. Each directory entry included a photograph, name, academic title, institutional affiliation, race, gender, research area(s), and a link to the faculty member’s webpage (see Figure S3 for an example screenshot). To assess the reliability of these classifications, an independent research assistant who was not involved in the original directory construction coded the race and gender of a random sample of 100 faculty members using each faculty member’s photograph, name, title, university, and website. Agreement between the independent research assistant’s coding and the original directory classifications was 92% for race and 100% for gender.

### **3.2 Control Directory**

Departments assigned to the *Directory Control* condition received a directory listing peer department names with hyperlinks to department homepages (see Figure S4 for an example). Peer departments were defined using the very same U.S. News and World Report ranking-based criteria as in the treatment condition.

#### **Section 4: Speaker-Directory Matching Protocol**

We assessed whether each of the people who filled one of the 23,455 STEM seminar-speaker slots under study in the 2024-25 academic year appeared in (or, in the control condition, *would* have appeared in) the relevant URM directory for that seminar’s organizer.

Because names appear in varied forms across data sources—differing in punctuation, middle initials, suffixes, and occasional misspellings—we employed a four-step matching protocol applied identically to Black and Hispanic speakers. First, all speaker and directory names were normalized by converting to lowercase, removing all non-alphabetic characters except spaces, collapsing consecutive whitespace, and trimming. Second, normalized speaker names were compared to normalized directory names via exact string equality. This “exact matching” process confirmed that 33 Black and 29 Hispanic seminar speaker appearances were filled by someone who appeared in a seminar organizer’s relevant directory. Third, for unmatched speakers, we computed the Jaro-Winkler distance (Jaro, 1989; Winkler, 1990) between every remaining speaker–directory name pair and flagged all pairs with a distance of 0.20 or below as candidates for manual review. Fourth, each candidate pair with a Jaro-Winkler distance of 0.20 or below was manually reviewed by two independent research assistants blind to condition who verified whether the two names plausibly referred to the same individual. Combining exact and fuzzy matching, this procedure identified a total of 41 unique Black and 46 unique Hispanic individuals who appeared in both a seminar lineup and a relevant organizer's directory.

## Section 5: Supplementary Figures

### Figure S1. Sample Recruitment Email

**From:** Seminar Support (seminar-support@wharton.upenn.edu)

**Sent:**

**To:**

**Subject:** your seminar series

Dear [Professor X/Seminar Organizer],

We're a non-profit at the University of Pennsylvania contacting you in your capacity as the [current chair of your department/organizer of your {seminar}].

To support seminar organizers, we have created a database of [scholars in your field who might/ departments in your field that you can search to] enhance the diversity of your academic seminar series, and you can find it [here](#).

This database has been personalized for your department.

Sincerely,



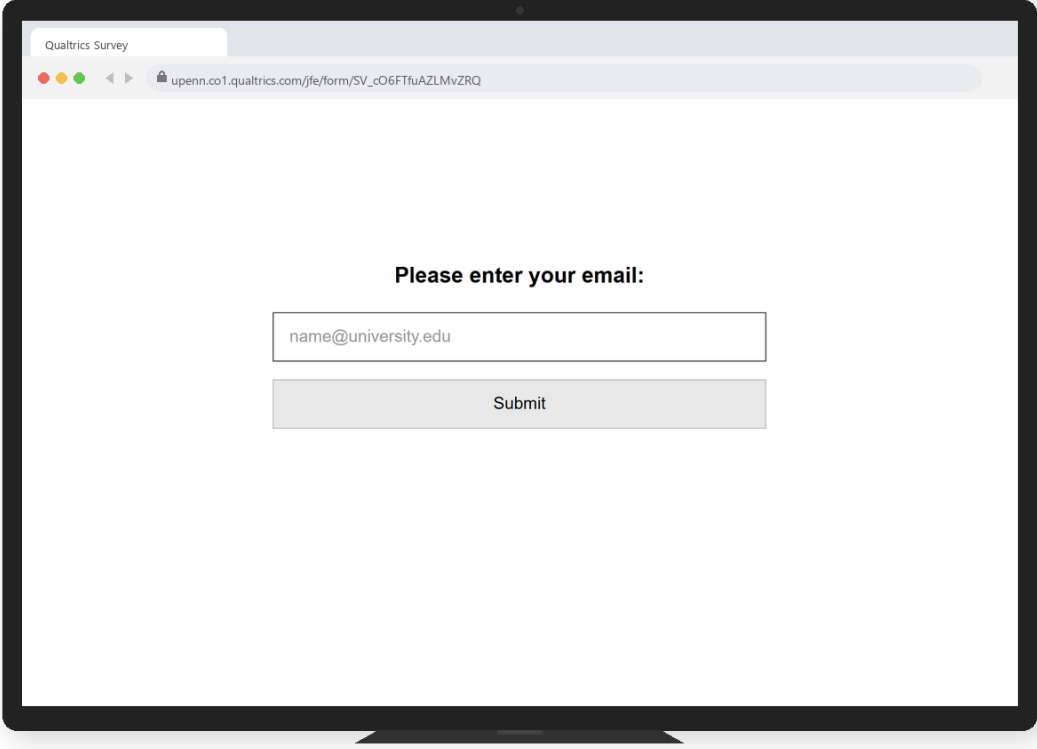
The Seminar Support Team  
The University of Pennsylvania

P.S. We'd be grateful if you would **please refrain from sharing this email with colleagues at other universities, as it has been tailor-made for you** and sharing it would jeopardize our goal of assessing its impact (though you are welcome to share it with others who co-organize your department's seminars).

*Notes:* Sample email sent to study participants. The email varied along two dimensions. Recipients were addressed either as "Seminar Organizer" or as "Professor X" depending on whether they were identified as their department's seminar organizer or department chair, respectively (yellow highlighting). Recipients were either told they were being contacted in their capacity as the "current chair of your department" or "organizer of your [name of seminar series inserted here]" depending on whether they were identified as their department's chair or seminar organizer, respectively (yellow highlighting). The email also contained a key treatment manipulation: (1) the *URM Directory Treatment* condition shared "a database of scholars in your field who might enhance the diversity of your academic seminar series" (red text), while (2) the *Directory Control* condition shared "a database of departments in your field that you can search to enhance the diversity of your academic seminar series" (blue text). Treatment emails included a hyperlink to a curated directory of underrepresented minority faculty from peer institutions (see Figure S3); control emails included a hyperlink to a list of peer department websites (see Figure S4). All emails were sent from a dedicated study email address during the week of June 24, 2024, with follow-up reminders sent the week of July 9, 2024 to non-responders. This email has been anonymized by removing the recipient's surname from the salutation and their email address. No highlighted or colorful text appeared in the actual study emails. All highlights and color have been added here to emphasize what was varied in our experiment.

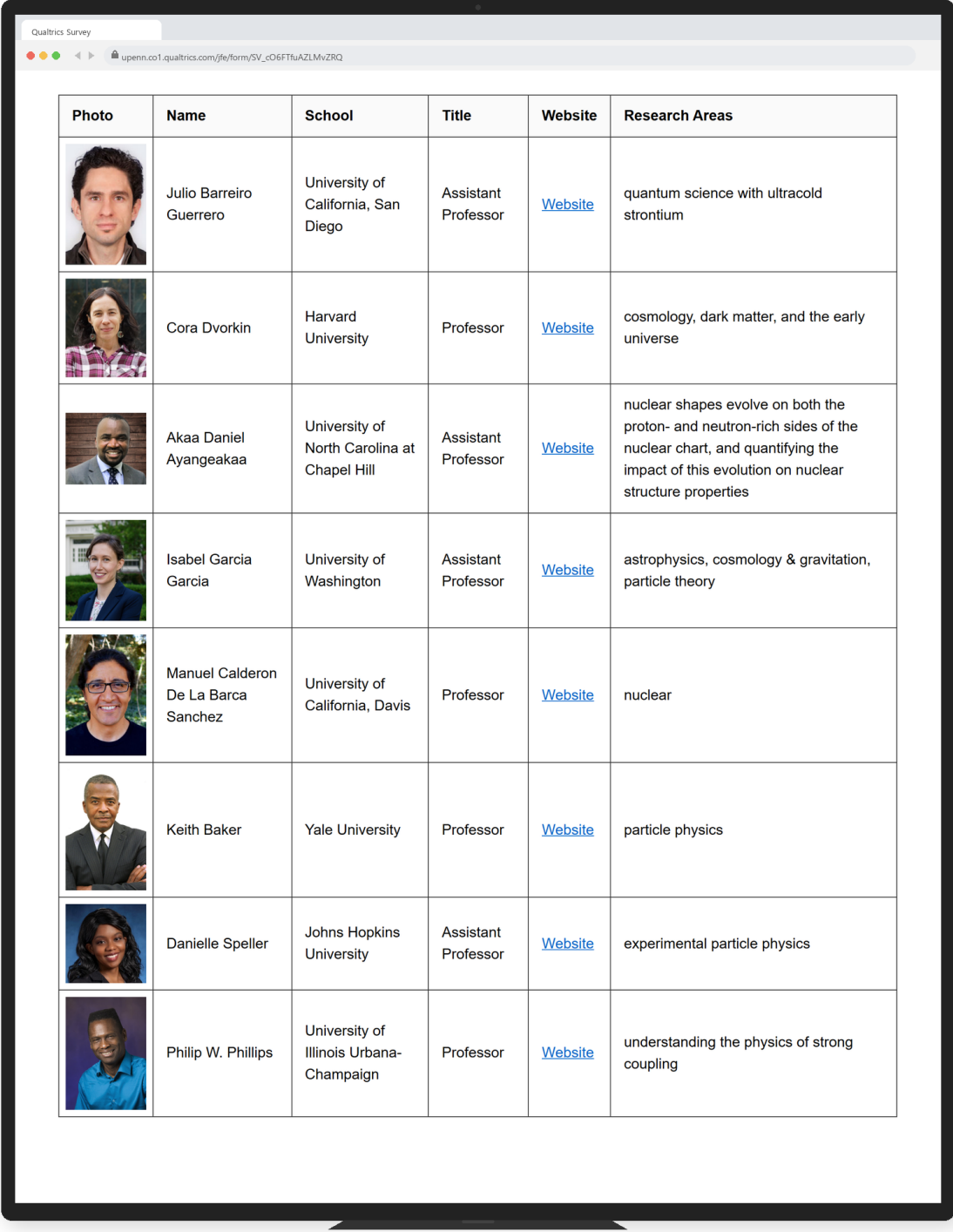
We sent at most 300 emails to participants per day in an attempt to avoid triggering institutional spam filters. Each person in our study received only one email regardless of how many seminars they oversaw. In cases where a department chair also served as a seminar organizer ( $n=8$ ), we treated them as the department chair and sent them only the department chair email, so the 1,019 seminar organizers and 567 department chairs represent mutually exclusive groups. To reduce sharing of study materials across institutions, recipients were also asked not to forward our email to colleagues at other universities.

**Figure S2. Directory Landing Page**

A screenshot of a web browser displaying a Qualtrics survey landing page. The browser's address bar shows the URL "upenn.co1.qualtrics.com/jfe/form/SV\_cO6FTfuAZLMvZRQ". The page content is centered and features the text "Please enter your email:" above a text input field. The input field contains the placeholder text "name@university.edu". Below the input field is a grey "Submit" button. The browser window has a dark border and standard OS window controls (red, yellow, green buttons) are visible in the top left corner.

*Notes:* The landing page displayed to both treatment and control condition study participants upon accessing their assigned directory link. Recipients were prompted to enter their university email address to verify their institutional affiliation before viewing their personalized directory. This verification step served to restrict directory access to intended recipients and reduce potential sharing of study materials across institutions.

**Figure S3. URM Directory Treatment Interface**

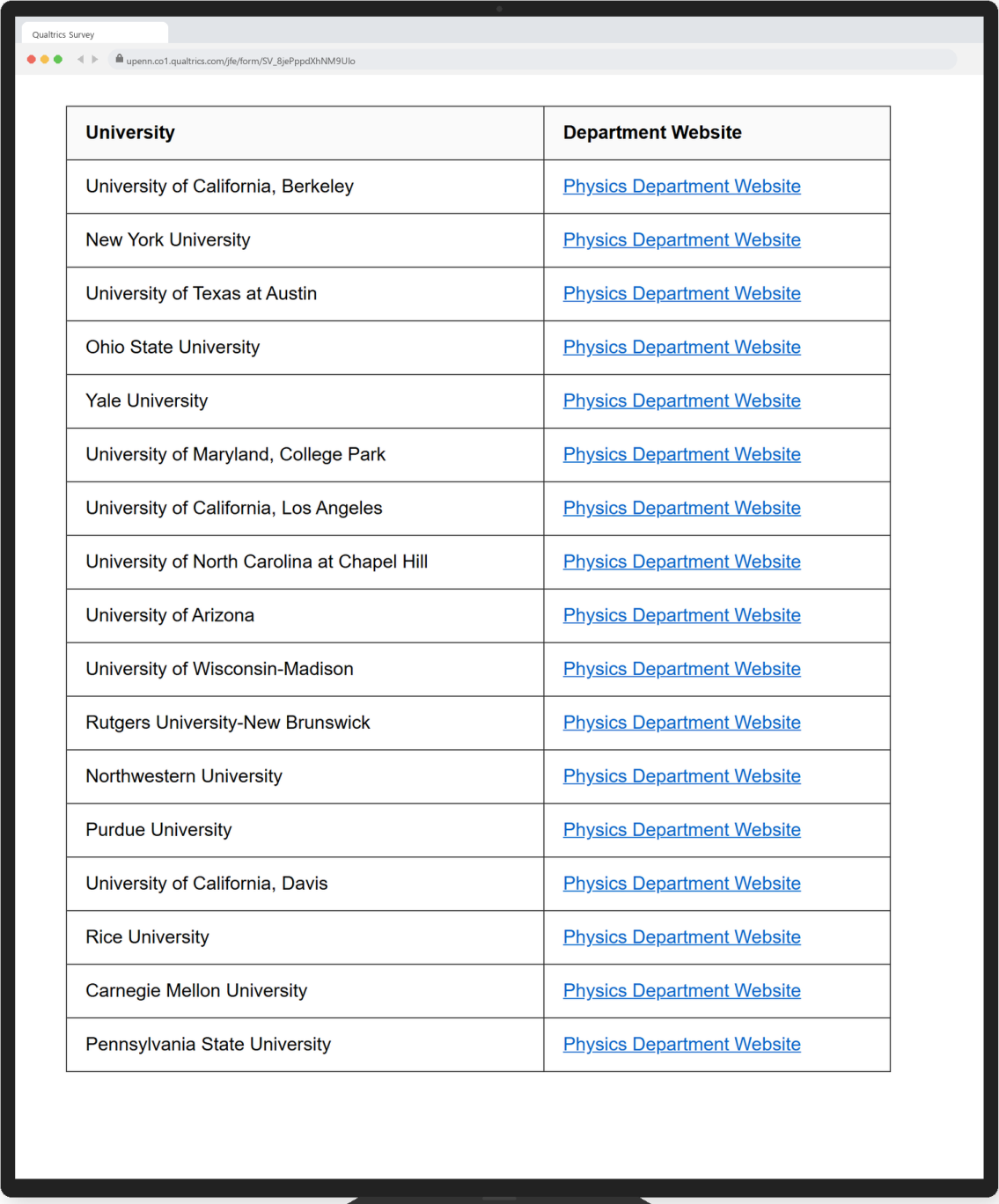


The image shows a tablet screen displaying a Qualtrics survey. The browser address bar shows the URL: upenn.co1.qualtrics.com/jfe/form/SV\_c06FTfuAZLMvZRQ. The survey content is a table with six columns: Photo, Name, School, Title, Website, and Research Areas. The table lists eight URM faculty members from various institutions, including the University of California, Harvard University, University of North Carolina at Chapel Hill, University of Washington, University of California, Davis, Yale University, Johns Hopkins University, and University of Illinois Urbana-Champaign. Each entry includes a headshot, name, institution, title, a link to their website, and their research areas.

| Photo                                                                               | Name                                | School                                      | Title               | Website                 | Research Areas                                                                                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------|---------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Julio Barreiro Guerrero             | University of California, San Diego         | Assistant Professor | <a href="#">Website</a> | quantum science with ultracold strontium                                                                                                                            |
|    | Cora Dvorkin                        | Harvard University                          | Professor           | <a href="#">Website</a> | cosmology, dark matter, and the early universe                                                                                                                      |
|    | Akaa Daniel Ayangeakaa              | University of North Carolina at Chapel Hill | Assistant Professor | <a href="#">Website</a> | nuclear shapes evolve on both the proton- and neutron-rich sides of the nuclear chart, and quantifying the impact of this evolution on nuclear structure properties |
|    | Isabel Garcia Garcia                | University of Washington                    | Assistant Professor | <a href="#">Website</a> | astrophysics, cosmology & gravitation, particle theory                                                                                                              |
|   | Manuel Calderon De La Barca Sanchez | University of California, Davis             | Professor           | <a href="#">Website</a> | nuclear                                                                                                                                                             |
|  | Keith Baker                         | Yale University                             | Professor           | <a href="#">Website</a> | particle physics                                                                                                                                                    |
|  | Danielle Speller                    | Johns Hopkins University                    | Assistant Professor | <a href="#">Website</a> | experimental particle physics                                                                                                                                       |
|  | Philip W. Phillips                  | University of Illinois Urbana-Champaign     | Professor           | <a href="#">Website</a> | understanding the physics of strong coupling                                                                                                                        |

*Notes:* Sample view of a portion of the personalized directory that would have been provided to study participants in MIT's physics department in the *URM Directory Treatment* condition. The directory displayed at least 20 underrepresented minority (URM) faculty members from peer departments (ranked within 20 positions of the recipient's department in the 2024 U.S. News & World Report rankings). Each entry included the faculty member's photograph, name, institution, academic title, website link, and research areas.

**Figure S4. Directory Control Interface**

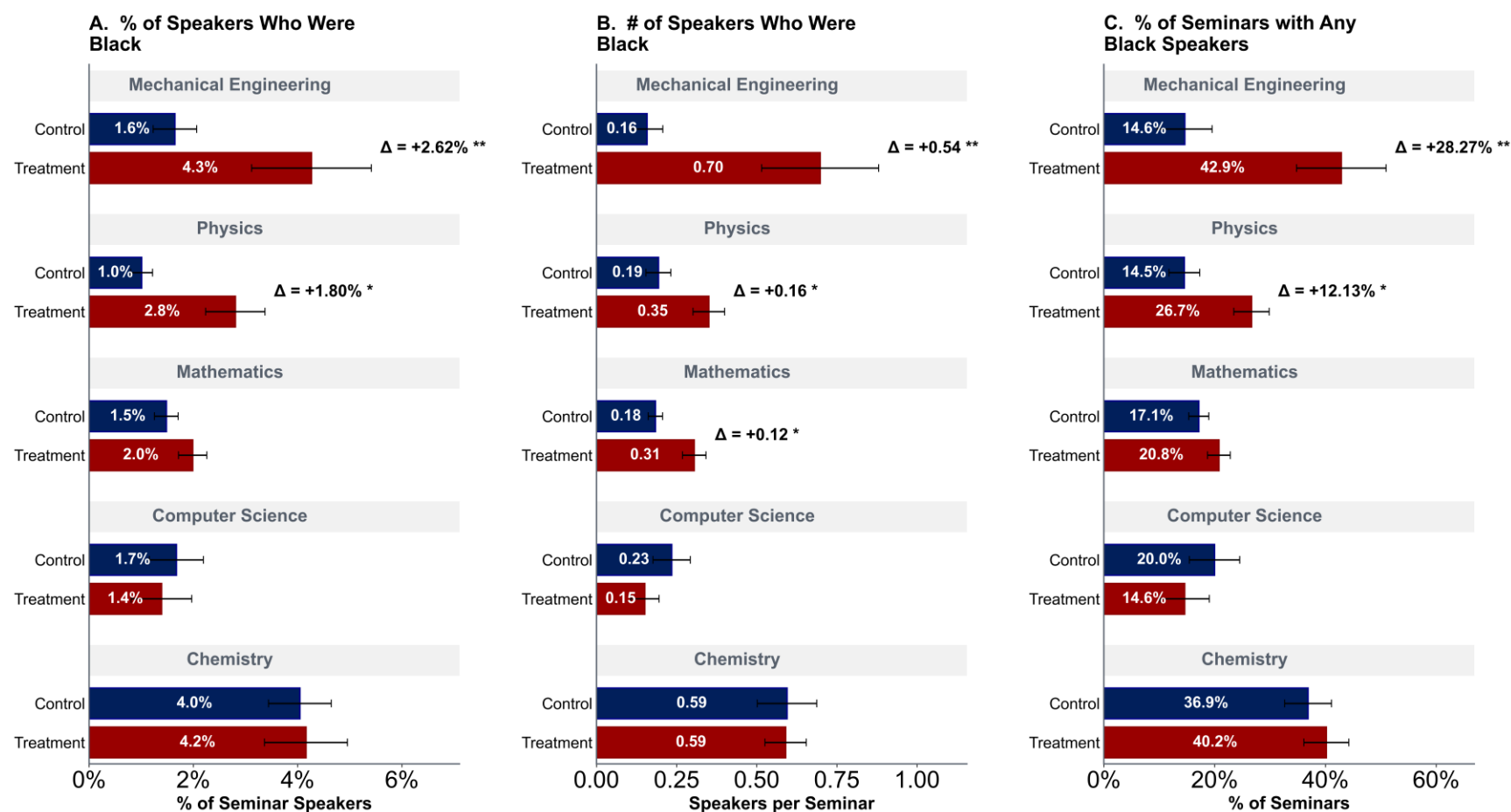


The image shows a screenshot of a web browser displaying a Qualtrics Survey. The browser's address bar shows the URL: upenn.co1.qualtrics.com/file/form/SV\_8jePpdXhNM9Ulo. The survey content is a table with two columns: 'University' and 'Department Website'. The table lists 16 universities, each with a corresponding link to its physics department website. The links are all identical: 'Physics Department Website'.

| University                                  | Department Website                         |
|---------------------------------------------|--------------------------------------------|
| University of California, Berkeley          | <a href="#">Physics Department Website</a> |
| New York University                         | <a href="#">Physics Department Website</a> |
| University of Texas at Austin               | <a href="#">Physics Department Website</a> |
| Ohio State University                       | <a href="#">Physics Department Website</a> |
| Yale University                             | <a href="#">Physics Department Website</a> |
| University of Maryland, College Park        | <a href="#">Physics Department Website</a> |
| University of California, Los Angeles       | <a href="#">Physics Department Website</a> |
| University of North Carolina at Chapel Hill | <a href="#">Physics Department Website</a> |
| University of Arizona                       | <a href="#">Physics Department Website</a> |
| University of Wisconsin-Madison             | <a href="#">Physics Department Website</a> |
| Rutgers University-New Brunswick            | <a href="#">Physics Department Website</a> |
| Northwestern University                     | <a href="#">Physics Department Website</a> |
| Purdue University                           | <a href="#">Physics Department Website</a> |
| University of California, Davis             | <a href="#">Physics Department Website</a> |
| Rice University                             | <a href="#">Physics Department Website</a> |
| Carnegie Mellon University                  | <a href="#">Physics Department Website</a> |
| Pennsylvania State University               | <a href="#">Physics Department Website</a> |

*Notes:* Sample view of a portion of the directory that would have been provided to study participants in Stanford University's physics department in the *Directory Control* condition. The directory listed peer departments (identical to those from which URM faculty were drawn in the *URM Directory Treatment* condition) with direct links to each department's website. This design required participants to independently search for and identify potential speakers, maintaining the typical processes associated with finding candidates.

**Figure S5. Treatment Effects on Black Speakers by Discipline**



*Notes:* This figure displays the treatment effects of Black speaker outcomes. Panel A displays the regression-adjusted percentage of seminar speakers who were Black, Panel B displays regression-adjusted number of speakers per seminar who were Black, and Panel C displays the regression-adjusted percentage of seminars with at least one Black speaker, each disaggregated by discipline. Blue bars represent seminars in the *Directory Control* condition (n=862), while red bars represent seminars in the *URM Directory Treatment* condition (n=824). Values are regression-adjusted predictions from ordinary least squares models run separately by discipline. All models include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted), and (2)

randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted). Models also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Error bars represent +/- 1 standard error. Standard errors are clustered at the seminar series' department level.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

## Section 6: Supplementary Tables

**Table S1. Analysis of Attrition in Academic Seminar Series Between the 2023-24 and 2024-25 Academic Years**

| Summary of How the Seminar Disappeared between 2023-24 and 2024-25                          | n          | %             | URM Directory Treatment | Directory Control  |
|---------------------------------------------------------------------------------------------|------------|---------------|-------------------------|--------------------|
| Maintained an active website but had no publicly listed speakers for 2024-25                | 43         | 22.1%         | 19 (17.6%)              | 24 (27.6%)         |
| Removed or restricted public access to information about speakers for 2024-25 <sup>a</sup>  | 87         | 44.6%         | 55 (50.9%)              | 32 (36.8%)         |
| Explicitly announced that it had been discontinued or was inactive for 2024-25 <sup>b</sup> | 65         | 33.3%         | 34 (31.5%)              | 31 (35.6%)         |
| <b>TOTAL</b>                                                                                | <b>195</b> | <b>100.0%</b> | <b>108 (100%)</b>       | <b>87 (100.0%)</b> |

*Notes:* This table reports the breakdown of 195 academic seminar series (10.4% of the original sample) that could not be tracked during the 2024–25 academic year outcome measurement period. The original experimental sample included 1,881 seminar series identified during the 2023–24 academic year across 146 R1 universities. Of these, 1,686 (89.6%) maintained publicly accessible speaker information in 2024-25 and constitute the analysis sample. This table categorizes the 195 missing seminars based on a systematic review of their department websites and online presence conducted between August 2024 and June 2025. The difference in attrition rates between treatment and control conditions was not statistically significant at either the department level (7.4% vs. 5.2%;  $B = 0.022$ ,  $SE = 0.021$ ,  $p = .284$ , HC3-robust) or the seminar level (11.6% vs. 9.2%;  $B = 0.024$ ,  $SE = 0.016$ ,  $p = .133$ , clustered at the department level); see Table S5 for full results. Treatment and control columns show n (% within condition). Seminars were selected on the basis of their 2023-24 characteristics, and these same seminar series were then tracked during the 2024-25 academic year regardless of whether their format changed.

<sup>a</sup> No public information includes seminars where the website was removed, locked behind university access, status uncertain, alternative links found without data, access restricted behind login, website errors, or otherwise no public information available.

<sup>b</sup> Discontinued for 2024–25 includes seminars explicitly marked as discontinued, with the last recorded activity in 2023 or Spring 2024.

**Table S2. Summary Statistics and Balance Checks (Analyzed Sample)**

| Characteristic                                      | Mean Across<br>Conditions | Mean in<br>Directory<br>Control | Mean in URM<br>Directory<br>Treatment | <i>p</i> -value<br>(RI) |
|-----------------------------------------------------|---------------------------|---------------------------------|---------------------------------------|-------------------------|
| Panel A: Department-Level Characteristics (N = 532) |                           |                                 |                                       |                         |
| # of Seminars Hosted by Department                  | 3.17 (3.50)               | 3.18 (3.52)                     | 3.16 (3.48)                           | 0.745                   |
| Chemistry Department                                | 22.93%                    | 19.93%                          | 26.05%                                | 0.082†                  |
| Computer Science Department                         | 15.41%                    | 17.71%                          | 13.03%                                | 0.131                   |
| Mathematics Department                              | 25.38%                    | 26.57%                          | 24.14%                                | 0.480                   |
| Mechanical Engineering Department                   | 12.41%                    | 11.81%                          | 13.03%                                | 0.596                   |
| Physics Department                                  | 23.87%                    | 23.99%                          | 23.75%                                | 0.994                   |
| Number of Faculty in Department                     | 34.05 (18.04)             | 32.92 (15.49)                   | 35.23 (20.31)                         | 0.116                   |
| % Black Faculty in Department                       | 1.63%                     | 1.43%                           | 1.84%                                 | 0.117                   |
| % Hispanic Faculty in Department                    | 2.42%                     | 2.26%                           | 2.59%                                 | 0.341                   |
| % Female Faculty in Department                      | 20.35%                    | 19.99%                          | 20.72%                                | 0.282                   |
| % Department Chairs who are Black                   | 2.26%                     | 3.33%                           | 1.15%                                 | 0.087†                  |
| % Department Chairs who are Hispanic                | 5.27%                     | 5.93%                           | 4.60%                                 | 0.465                   |
| % Department Chairs who are Female                  | 17.89%                    | 18.52%                          | 17.24%                                | 0.677                   |
| Joint F-test (RI p-value)                           | <i>F</i> (12, 512) = 1.66 |                                 |                                       | 0.108                   |
| Panel B: Seminar-Level Characteristics (N = 1,686)  |                           |                                 |                                       |                         |
| Number of Email Recipients                          | 4.86 (4.46)               | 4.68 (4.25)                     | 5.05 (4.67)                           | 0.516                   |
| % Seminar Organizers who are Black                  | 4.70%                     | 5.65%                           | 3.74%                                 | 0.519                   |
| % Seminar Organizers who are Hispanic               | 7.77%                     | 7.14%                           | 8.40%                                 | 0.634                   |
| % Seminar Organizers who are Female                 | 43.59%                    | 44.98%                          | 42.18%                                | 0.573                   |
| Joint F-test (RI p-value)                           | <i>F</i> (4, 1344) = 0.32 |                                 |                                       | 0.925                   |

*Notes:* This table reports balance checks for our final analytic sample including 532 departments and 1,686 seminars. Panel A reports department-level characteristics with each department weighted equally. Panel B reports seminar-level characteristics. Each row reports condition-specific means for the pre-treatment characteristic listed in the row label. Following Athey and Imbens (2017), the p-value column reports randomization inference (RI) p-values from 5,000 within-strata permutations of the treatment assignment, based on the absolute t-statistic from an OLS regression of the characteristic on the treatment indicator with randomization-strata fixed effects. Panel A uses HC3-robust standard errors; Panel B clusters standard errors at the department level. The joint F-test at the bottom of each panel tests the null that all listed covariates are jointly unrelated to treatment assignment. Randomization inference follows Hansen and Bowers (2008) and preserves the original stratified randomization design. Note that the number of seminars used in stratification was based on the 1,881 seminars in our original sample and how many seminars a department hosted in the 2023–24 academic year, and was not based on the number of seminars a department hosted that continued into the 2024–25 academic year (this final analysis sample included 195 fewer seminars total). We stratified department randomization based on the number of seminars a department hosted, creating seven strata: 1 seminar (n = 280), 2-3 seminars (n = 280), 4-5 seminars (n = 300), 6-7 seminars (n = 278), 8-11 seminars (n = 298), 12-17 seminars (n = 318), and 18-28 seminars (n = 127). Within each stratum, departments were randomly assigned with equal probability to each condition.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

Directory engagement. The average number of email link clicks per seminar was significantly higher in the URM Directory Treatment condition: participants clicked 0.74 times per seminar in the URM Directory Treatment condition compared to 0.63 times in the Directory Control condition,  $t(1684) = 1.99$ ,  $p = .047$ .

**Table S3. Characteristics of Speakers in STEM Seminars in the 2024–25 Academic Year, by Race/Ethnicity**

|                                                     | All            | White          | Asian         | Hispanic    | Black       |
|-----------------------------------------------------|----------------|----------------|---------------|-------------|-------------|
| <i>N</i> appearances                                | 23,455         | 14,620         | 7,179         | 1,127       | 523         |
| <i>N</i> unique speakers                            | 17,712         | 10,999         | 5,391         | 887         | 430         |
| <b>Panel A: Speaker's Gender</b>                    |                |                |               |             |             |
| Woman                                               | 3,986 (17.0%)  | 2,578 (17.6%)  | 1,118 (15.6%) | 198 (17.6%) | 90 (17.2%)  |
| Man                                                 | 19,469 (83.0%) | 12,042 (82.4%) | 6,061 (84.4%) | 929 (82.4%) | 433 (82.8%) |
| <b>Panel B: Speaker's Academic Rank</b>             |                |                |               |             |             |
| Professor                                           | 6,605 (28.2%)  | 4,731 (32.4%)  | 1,507 (21.0%) | 283 (25.1%) | 84 (16.1%)  |
| Associate Professor                                 | 2,109 (9.0%)   | 1,338 (9.2%)   | 633 (8.8%)    | 84 (7.5%)   | 53 (10.1%)  |
| Assistant Professor                                 | 4,441 (18.9%)  | 2,413 (16.5%)  | 1,708 (23.8%) | 209 (18.5%) | 109 (20.8%) |
| Post-doc                                            | 2,622 (11.2%)  | 1,473 (10.1%)  | 956 (13.3%)   | 135 (12.0%) | 56 (10.7%)  |
| Graduate Student                                    | 3,080 (13.1%)  | 1,594 (10.9%)  | 1,204 (16.8%) | 182 (16.1%) | 99 (18.9%)  |
| Non-faculty                                         | 4,598 (19.6%)  | 3,071 (21.0%)  | 1,171 (16.3%) | 234 (20.8%) | 122 (23.3%) |
| <b>Panel C: Speaker's Institutional Affiliation</b> |                |                |               |             |             |
| Higher-ranked (external)                            | 6,891 (29.4%)  | 4,086 (27.9%)  | 2,359 (32.9%) | 290 (25.7%) | 156 (29.8%) |
| Lower-ranked (external)                             | 4,045 (17.2%)  | 2,492 (17.0%)  | 1,297 (18.1%) | 183 (16.2%) | 73 (14.0%)  |
| Host institution                                    | 5,085 (21.7%)  | 3,100 (21.2%)  | 1,583 (22.1%) | 250 (22.2%) | 150 (28.7%) |
| Not in U.S. News rankings                           | 7,434 (31.7%)  | 4,942 (33.8%)  | 1,940 (27.0%) | 404 (35.8%) | 144 (27.5%) |
| At Institution in Peer Set                          | 4,840 (20.6%)  | 2,874 (19.7%)  | 1,673 (23.3%) | 192 (17.0%) | 101 (19.3%) |
| Not at Institution in Peer Set                      | 18,615 (79.4%) | 11,746 (80.3%) | 5,506 (76.7%) | 935 (83.0%) | 422 (80.7%) |

*Notes:* This table describes the characteristics of the speakers who appeared across 1,686 STEM seminars in our analytic sample during the 2024–25 academic year, broken down by speaker race/ethnicity. The unit of observation is a speaker appearance (a speaker may appear in multiple seminars and would then be counted repeatedly here). Entries show the count and column percentage within each racial/ethnic group. Academic rank is as reported in public institutional profiles. Institutional affiliation categorizes speakers relative to the host department's 2024 U.S. News & World Report discipline-specific ranking. Higher-ranked (external) includes speakers from institutions ranked above the host; lower-ranked (external) includes those ranked below. Host institution includes speakers whose affiliation matches the seminar's host university. Unranked includes speakers from international, industry, or other affiliations not in our ranking data. At Institution in Peer Set includes speakers whose institutional affiliation is one of the approximately 40 peer departments ranked within +/- 20 positions of the host department—the same set of institutions eligible for inclusion in the URM directory (see Table 3 in the main text). Not at Institution in Peer Set includes all remaining speakers (those at the host institution, at unranked institutions, or at ranked institutions outside the peer set). Race/ethnicity and gender were coded using a multi-method approach combining facial recognition, name-based classification, and manual verification (see Methods section for details).

**Table S4. Summary Statistics and Balance Checks (Full Randomized Sample)**

| Characteristic                                      | Mean Across<br>Conditions | Mean in<br>Directory<br>Control | Mean in URM<br>Directory<br>Treatment | p-value<br>(RI) |
|-----------------------------------------------------|---------------------------|---------------------------------|---------------------------------------|-----------------|
| Panel A: Department-Level Characteristics (N = 568) |                           |                                 |                                       |                 |
| # of Seminars Hosted by Department                  | 3.31 (3.74)               | 3.32 (3.74)                     | 3.30 (3.75)                           | 0.885           |
| Chemistry Department                                | 22.54%                    | 19.58%                          | 25.53%                                | 0.097†          |
| Computer Science Department                         | 16.55%                    | 18.18%                          | 14.89%                                | 0.298           |
| Mathematics Department                              | 24.12%                    | 25.52%                          | 22.70%                                | 0.423           |
| Mechanical Engineering Department                   | 13.91%                    | 13.29%                          | 14.54%                                | 0.639           |
| Physics Department                                  | 22.89%                    | 23.43%                          | 22.34%                                | 0.747           |
| Number of Faculty in Department                     | 33.57 (17.82)             | 32.42 (15.36)                   | 34.73 (19.97)                         | 0.102           |
| % Black Faculty in Department                       | 1.61%                     | 1.41%                           | 1.82%                                 | 0.099†          |
| % Hispanic Faculty in Department                    | 2.50%                     | 2.30%                           | 2.71%                                 | 0.262           |
| % Female Faculty in Department                      | 20.19%                    | 19.82%                          | 20.56%                                | 0.256           |
| % Department Chairs who are Black                   | 2.12%                     | 3.16%                           | 1.06%                                 | 0.076†          |
| % Department Chairs who are Hispanic                | 5.29%                     | 5.61%                           | 4.96%                                 | 0.773           |
| % Department Chairs who are Female                  | 17.28%                    | 17.89%                          | 16.67%                                | 0.664           |
| Joint F-test (RI p-value)                           | F(12, 549) = 1.56         |                                 |                                       | 0.154           |
| Panel B: Seminar-Level Characteristics (N = 1,881)  |                           |                                 |                                       |                 |
| Number of Email Recipients                          | 4.82 (4.41)               | 4.65 (4.22)                     | 4.99 (4.59)                           | 0.498           |
| % Seminar Organizers who are Black                  | 4.80%                     | 5.89%                           | 3.73%                                 | 0.555           |
| % Seminar Organizers who are Hispanic               | 7.74%                     | 7.49%                           | 7.99%                                 | 0.885           |
| % Seminar Organizers who are Female                 | 43.35%                    | 44.32%                          | 42.38%                                | 0.638           |
| Joint F-test (RI p-value)                           | F(4, 1505) = 0.30         |                                 |                                       | 0.932           |

*Notes:* This table reports balance checks for the full randomized sample (568 departments, 1,881 seminars), including the 36 departments (195 seminars) that attrited from the final analytic sample. Panel A reports department-level characteristics with each department weighted equally. Panel B reports seminar-level characteristics. Each row reports condition-specific means for the pre-treatment characteristic listed in the row label. Following Athey and Imbens (2017), the p-value column reports randomization inference (RI) p-values from 5,000 within-strata permutations of the treatment assignment, based on the absolute t-statistic from an OLS regression of the characteristic on the treatment indicator with randomization-strata fixed effects. Panel A uses HC3-robust standard errors; Panel B clusters standard errors at the department level. The joint F-test at the bottom of each panel tests the null that all listed covariates are jointly unrelated to treatment assignment. Email recipients were distributed across disciplines as follows: 42.2% worked in Mathematics departments, 20.1% worked in Physics departments, 16.1% worked in Chemistry departments, 13.0% worked in Computer Science departments, and 8.7% worked in Mechanical Engineering departments. Randomization inference follows Hansen and Bowers (2008) and preserves the original stratified randomization design.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

**Table S5. Tests for Differential Attrition of Seminars and Departments across Conditions**

|                          | <b>Did All Seminars in the<br/>Department Disappear between<br/>Treatment Assignment and<br/>Outcome Measurement?<br/>(1)</b> | <b>Did the Seminar Disappear<br/>between Treatment<br/>Assignment and Outcome<br/>Measurement?<br/>(2)</b> |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| URM Directory Treatment  | 0.0220<br>(0.0205)                                                                                                            | 0.0242<br>(0.0161)                                                                                         |
| p-value                  | 0.284                                                                                                                         | 0.133                                                                                                      |
| Control attrition rate   | 5.2%                                                                                                                          | 9.2%                                                                                                       |
| Treatment attrition rate | 7.4%                                                                                                                          | 11.6%                                                                                                      |
| N                        | 568                                                                                                                           | 1,881                                                                                                      |

*Notes:* This table tests for differential attrition by treatment condition. Column (1) reports a department-level OLS regression where the outcome variable is a binary indicator for whether the department in question dropped out of the analytic sample. The primary predictor is a *URM Directory Treatment* indicator. The regression includes HC3-robust standard errors (N = 568 departments). Column (2) reports a seminar-level OLS regression where the outcome variable is a binary indicator for whether the seminar series in question dropped out of the analytic sample. The primary predictor is a *URM Directory Treatment* indicator. The regression includes standard errors clustered at the department level (N = 1,881 seminars).

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

**Table S6. Intent-to-Treat (ITT) Analysis with Imputation of Outcomes in Unobserved Seminars**

|                                                                    | DV: % of AY24-25<br>Seminars Speakers Who<br>Were Black |          | DV: Number of AY24-25<br>Seminars Speakers Who<br>Were Black |          | DV: Did AY24-25 Seminar<br>Have at Least One Black<br>Speaker? |           |
|--------------------------------------------------------------------|---------------------------------------------------------|----------|--------------------------------------------------------------|----------|----------------------------------------------------------------|-----------|
|                                                                    | (1)                                                     | (2)      | (3)                                                          | (4)      | (5)                                                            | (6)       |
| URM Directory<br>Treatment                                         | 0.5908*                                                 | 0.5920*  | 0.0657†                                                      | 0.0656*  | 0.0439*                                                        | 0.0438*   |
|                                                                    | (0.2742)                                                | (0.2633) | (0.0347)                                                     | (0.0332) | (0.0201)                                                       | (0.0202)  |
| # of Faculty in Dept.<br>of Seminar                                |                                                         | -0.0147* |                                                              | -0.0019* |                                                                | -0.0011†  |
|                                                                    |                                                         | (0.0060) |                                                              | (0.0009) |                                                                | (0.0006)  |
| % of Faculty in Dept. of<br>Seminars Who Are Female                |                                                         | 2.2260   |                                                              | 0.3049   |                                                                | 0.1956    |
|                                                                    |                                                         | (1.8111) |                                                              | (0.2540) |                                                                | (0.1512)  |
| % of Faculty in Dept. of<br>Seminars Who Are URMs                  |                                                         | 6.1283   |                                                              | 0.4193   |                                                                | 0.2409    |
|                                                                    |                                                         | (4.2155) |                                                              | (0.5193) |                                                                | (0.2810)  |
| U.S. News Ranking of<br>Dept. of Seminar                           |                                                         | 0.0094   |                                                              | 0.0012   |                                                                | -0.0000   |
|                                                                    |                                                         | (0.0061) |                                                              | (0.0008) |                                                                | (0.0004)  |
| # of Faculty Displayed in<br>URM Directory for Dept.<br>of Seminar |                                                         | 0.0553*  |                                                              | 0.0086** |                                                                | 0.0051**  |
|                                                                    |                                                         | (0.0264) |                                                              | (0.0033) |                                                                | (0.0019)  |
| # of Peer Departments with<br>20+ URM Faculty                      |                                                         | 0.1725   |                                                              | 0.0083   |                                                                | 0.0051    |
|                                                                    |                                                         | (0.1402) |                                                              | (0.0114) |                                                                | (0.0061)  |
| # of Dept Reps Who<br>Received Treatment Email                     |                                                         | 0.0775** |                                                              | 0.0128** |                                                                | 0.0093*** |
|                                                                    |                                                         | (0.0290) |                                                              | (0.0040) |                                                                | (0.0025)  |
| Observations                                                       | 1,881                                                   | 1,881    | 1,881                                                        | 1,881    | 1,881                                                          | 1,881     |
| Adjusted R <sup>2</sup>                                            | 0.031                                                   | 0.037    | 0.051                                                        | 0.062    | 0.044                                                          | 0.054     |

*Notes:* This table re-estimates treatment effects on the full randomized sample of 1,881 seminars, imputing zeros for Black-speaker outcomes in the 195 seminars that were not observable in 2024–25. This conservative specification assumes that unobserved seminars featured no Black speakers. Models 1–2 predict the percentage of Black speakers in a given seminar series in the 2024–25 academic year. Models 3–4 predict the number of Black speakers in a given seminar series. Models 5–6 predict whether a given seminar series featured at least one Black speaker. Models 1, 3, and 5 include fixed effects for (1) the seminar’s discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series’ organizers (here we include nine fixed effects for different email batches sent between June 24–28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023–24 academic year (1 seminar series, 2–3 seminar series, 4–5 seminar series, 6–7 seminar series, 8–11 seminar series, 12–17 seminar series; an indicator for hosting 18–28 seminar series is omitted). Models 2, 4, and 6 include the same fixed effects but also include controls for (1) the total number of faculty in the seminar’s host department in summer 2024, (2) the fraction of faculty in the seminar’s host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar’s host department who were women in summer 2024, (4) the seminar’s host department’s U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university’s overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series’ department level, are shown in parentheses.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

**Table S7. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation in STEM Seminars in the 2024-25 Academic Year by Semester**

|                                                              | DV: % of Seminar Speakers Who Were Black |                       |                            |                      | DV: Number of Seminar Speakers Who Were Black |                       |                            |                      | DV: Did Seminar Have at Least One Black Speaker? |                       |                            |                      |
|--------------------------------------------------------------|------------------------------------------|-----------------------|----------------------------|----------------------|-----------------------------------------------|-----------------------|----------------------------|----------------------|--------------------------------------------------|-----------------------|----------------------------|----------------------|
|                                                              | <i>Fall '24 Semester</i>                 |                       | <i>Spring '25 Semester</i> |                      | <i>Fall '24 Semester</i>                      |                       | <i>Spring '25 Semester</i> |                      | <i>Fall '24 Semester</i>                         |                       | <i>Spring '25 Semester</i> |                      |
|                                                              | (1)                                      | (2)                   | (3)                        | (4)                  | (5)                                           | (6)                   | (7)                        | (8)                  | (9)                                              | (10)                  | (11)                       | (12)                 |
| URM Directory Treatment                                      | 0.3946<br>(0.2960)                       | 0.4147<br>(0.2873)    | 0.8694*<br>(0.3878)        | 0.8913*<br>(0.3755)  | 0.0458*<br>(0.0186)                           | 0.0459*<br>(0.0183)   | 0.0357<br>(0.0291)         | 0.0348<br>(0.0280)   | 0.0373*<br>(0.0148)                              | 0.0380*<br>(0.0151)   | 0.0271<br>(0.0190)         | 0.0271<br>(0.0188)   |
| # of Faculty in Dept. of Seminar                             |                                          | -0.0142*<br>(0.0072)  |                            | -0.0163†<br>(0.0088) |                                               | -0.0006<br>(0.0005)   |                            | -0.0014*<br>(0.0007) |                                                  | -0.0006<br>(0.0005)   |                            | -0.0010†<br>(0.0005) |
| % of Faculty in Dept. of Seminar Who Are Female              |                                          | 2.0216<br>(1.7778)    |                            | 2.4117<br>(2.6715)   |                                               | 0.0268<br>(0.1303)    |                            | 0.2269<br>(0.2138)   |                                                  | 0.0026<br>(0.1081)    |                            | 0.1910<br>(0.1486)   |
| % of Faculty in Dept. of Seminar Who Are URM                 |                                          | 5.8266<br>(4.2557)    |                            | 3.4763<br>(5.6748)   |                                               | 0.3594<br>(0.2998)    |                            | 0.1469<br>(0.3949)   |                                                  | 0.3391<br>(0.2521)    |                            | 0.0824<br>(0.2594)   |
| U.S. News Ranking of Dept. of Seminar                        |                                          | 0.0226**<br>(0.0074)  |                            | 0.0055<br>(0.0084)   |                                               | 0.0012**<br>(0.0004)  |                            | 0.0003<br>(0.0007)   |                                                  | 0.0007*<br>(0.0003)   |                            | -0.0000<br>(0.0004)  |
| # of Faculty Displayed in URM Directory for Dept. of Seminar |                                          | 0.1232***<br>(0.0312) |                            | 0.0234<br>(0.0351)   |                                               | 0.0075***<br>(0.0018) |                            | 0.0016<br>(0.0027)   |                                                  | 0.0059***<br>(0.0014) |                            | 0.0015<br>(0.0018)   |
| # of Peer Departments with 20+ URM Faculty                   |                                          | 0.2354<br>(0.1881)    |                            | 0.1353<br>(0.1830)   |                                               | 0.0034<br>(0.0062)    |                            | 0.0088<br>(0.0103)   |                                                  | 0.0026<br>(0.0054)    |                            | 0.0053<br>(0.0065)   |
| # of Dept Reps Who Received Treatment Email                  |                                          | 0.0837*<br>(0.0354)   |                            | 0.0727†<br>(0.0441)  |                                               | 0.0047*<br>(0.0024)   |                            | 0.0090**<br>(0.0032) |                                                  | 0.0040*<br>(0.0019)   |                            | 0.0069**<br>(0.0024) |
| Observations                                                 | 1,686                                    | 1,686                 | 1,686                      | 1,686                | 1,686                                         | 1,686                 | 1,686                      | 1,686                | 1,686                                            | 1,686                 | 1,686                      | 1,686                |
| Adjusted R <sup>2</sup>                                      | 0.015                                    | 0.033                 | 0.013                      | 0.012                | 0.023                                         | 0.036                 | 0.028                      | 0.037                | 0.020                                            | 0.032                 | 0.022                      | 0.025                |

*Notes:* This table reports the results of twelve OLS regressions estimating the effect of random assignment to the URM Directory Treatment condition on Black speaker representation, disaggregated by semester. Models 1–2 predict the percentage of Black speakers in the fall 2024 semester; Models 3–4 predict the percentage in the spring 2025 semester; Models 5–6 predict the number of Black speakers in the fall 2024 semester; Models 7–8 predict the number of Black speakers in the spring 2025 semester; Models 9–10 predict whether a seminar featured at least one Black speaker in the fall 2024 semester; Models 11–12 predict whether a seminar featured at least one Black speaker in the spring 2025 semester. Within each pair of regressions, odd-numbered models include only fixed effects for (1) the seminar’s discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series’ organizers (here we include nine fixed effects for different email batches sent between June 24–28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023–24 academic year (1 seminar series, 2–3 seminar series, 4–5 seminar series, 6–7 seminar series, 8–11 seminar series, 12–17 seminar series; an indicator for hosting 18–28 seminar series is omitted). Even-numbered models include the same fixed effects but also include controls for (1) the total number of

faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URM in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. For seminars with no speakers in a given semester (Fall: 508; Spring: 308), percentage outcomes are coded as zero. Results are substantively unchanged when these seminars are excluded. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses. For consistency across institutions, we define "Fall semester" as seminars occurring between August 1 and December 31, 2024, and "Spring semester" as seminars occurring between January 1 and June 30, 2025, regardless of whether the institution uses a semester, trimester, or quarter system. † $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$

As expected given the null effects overall, semester-by-semester analyses of overall URM speaker and Hispanic speaker representation revealed no significant effects (all  $p$ 's  $> .10$ ; see Extended Tables E3–E4 in the replication repository).

## 6.1 Confidence Sets for Ranks of Discipline-Specific Treatment Effects

We applied the confidence sets for ranks procedure developed by Mogstad et al. (2024) to formally characterize the ranking of discipline-specific treatment effects. Marginal, one-sided, 95% confidence intervals were constructed for the rank of each discipline's treatment effect based on the Treatment x Discipline interaction coefficients reported in Table S8 below. When estimating the impact of our treatment on the percentage of Black speakers in a given seminar over the year post-intervention, Mechanical Engineering and Physics are the only disciplines whose 95% confidence set for rank includes 1 (i.e., the top position). When estimating the impact of our treatment on the number of Black speakers in a given seminar over the year post-intervention and the probability of a seminar featuring any Black speaker in the year post-intervention, Mechanical Engineering is definitively the top-ranked discipline. Table S9 reports these confidence sets.

**Table S8. Linear Regressions Estimating Heterogeneity in Treatment Effects on Black Speaker Representation by Academic Discipline in the 2024-25 Academic Year**

|                                                     | DV: % of AY24-25<br>Seminar Speakers Who<br>Were Black |                                   | DV: Number of AY24-25<br>Seminar Speakers Who<br>Were Black |                                  | DV: Did AY24-25 Seminar<br>Have at Least One Black<br>Speaker? |                                  |
|-----------------------------------------------------|--------------------------------------------------------|-----------------------------------|-------------------------------------------------------------|----------------------------------|----------------------------------------------------------------|----------------------------------|
|                                                     | (1)                                                    | (2)                               | (3)                                                         | (4)                              | (5)                                                            | (6)                              |
| URM Directory Treatment x<br>Mechanical Engineering | 3.8488***<br>(1.1653)                                  | 3.8227**<br>(1.1909)              | 0.5813**<br>(0.1878)                                        | 0.5740**<br>(0.1814)             | 0.3210**<br>(0.0984)                                           | 0.3375***<br>(0.1010)            |
| URM Directory Treatment x<br>Physics                | 1.6506**<br>(0.6309)                                   | 1.6345*<br>(0.6495)               | 0.1451*<br>(0.0655)                                         | 0.1433*<br>(0.0675)              | 0.0924*<br>(0.0446)                                            | 0.0932*<br>(0.0461)              |
| URM Directory Treatment x<br>Chemistry              | 0.4932<br>(1.0210)                                     | 0.4534<br>(0.9276)                | -0.0500<br>(0.1272)                                         | -0.0522<br>(0.1159)              | 0.0607<br>(0.0667)                                             | 0.0634<br>(0.0627)               |
| URM Directory Treatment x<br>Mathematics            | 0.2254<br>(0.3872)                                     | 0.1948<br>(0.3806)                | 0.0767<br>(0.0499)                                          | 0.0686<br>(0.0490)               | 0.0270<br>(0.0292)                                             | 0.0203<br>(0.0283)               |
| URM Directory Treatment x<br>Computer Science       | -0.2053<br>(0.8615)                                    | 0.1502<br>(0.8474)                | -0.0727<br>(0.0890)                                         | -0.0272<br>(0.0881)              | -0.0379<br>(0.0762)                                            | -0.0253<br>(0.0732)              |
| Mechanical Engineering                              | -0.2903<br>(0.7289)                                    | -12.1672 <sup>†</sup><br>(7.1803) | -0.0595<br>(0.0931)                                         | -1.0360 <sup>†</sup><br>(0.5945) | 0.0149<br>(0.0740)                                             | -0.6639*<br>(0.3126)             |
| Physics                                             | -0.0403<br>(0.4938)                                    | -12.1809 <sup>†</sup><br>(7.2239) | 0.0806<br>(0.0681)                                          | -0.8982<br>(0.5923)              | 0.0870*<br>(0.0430)                                            | -0.5577 <sup>†</sup><br>(0.3122) |
| Chemistry                                           | 2.6141**<br>(0.8724)                                   | -8.9020<br>(6.9897)               | 0.4815***<br>(0.1201)                                       | -0.4253<br>(0.5717)              | 0.2739***<br>(0.0621)                                          | -0.3316<br>(0.3041)              |
| Mathematics                                         | 0.8435*<br>(0.4236)                                    | -11.2574<br>(7.0025)              | 0.1399*<br>(0.0599)                                         | -0.8339<br>(0.5768)              | 0.1346**<br>(0.0429)                                           | -0.4850<br>(0.3037)              |
| Computer Science                                    | 0.4117<br>(0.7086)                                     | -11.1433<br>(7.1328)              | 0.0739<br>(0.0925)                                          | -0.8383<br>(0.5747)              | 0.0989<br>(0.0654)                                             | -0.5221 <sup>†</sup><br>(0.3064) |
| # of Faculty in Dept.<br>of Seminar                 |                                                        | -0.0146*<br>(0.0067)              |                                                             | -0.0020*<br>(0.0010)             |                                                                | -0.0010<br>(0.0006)              |
| % of Faculty in Dept. of<br>Seminar Who Are Female  |                                                        | 2.3354<br>(1.9308)                |                                                             | 0.3012<br>(0.2728)               |                                                                | 0.1921<br>(0.1638)               |
| % of Faculty in Dept. of                            |                                                        | 5.7742                            |                                                             | 0.4434                           |                                                                | 0.1875                           |

|                            |                     |          |           |
|----------------------------|---------------------|----------|-----------|
| Seminar Who Are URM        | (4.6080)            | (0.5777) | (0.3057)  |
| U.S. News Ranking of       | 0.0112 <sup>†</sup> | 0.0014   | 0.0001    |
| Dept. of Seminar           | (0.0065)            | (0.0009) | (0.0004)  |
| # of Faculty Displayed in  | 0.0628*             | 0.0094** | 0.0059**  |
| URM Directory for Dept.    | (0.0286)            | (0.0035) | (0.0020)  |
| of Seminar                 |                     |          |           |
| # of Peer Departments with | 0.2275              | 0.0133   | 0.0084    |
| 20+ URM Faculty            | (0.1656)            | (0.0133) | (0.0070)  |
| # of Dept Reps Who         | 0.0870**            | 0.0136** | 0.0101*** |
| Received Treatment Email   | (0.0311)            | (0.0044) | (0.0027)  |
| Observations               | 1,686               | 1,686    | 1,686     |
| Adjusted R <sup>2</sup>    | 0.144               | 0.151    | 0.216     |

*Notes:* This table reports the results of six OLS regressions estimating treatment-by-discipline interaction effects on Black speaker representation, where the primary predictors in each model are the interactions between treatment assignment and academic discipline indicators. All models suppress the intercept and include indicators for all five disciplines studied (mathematics, physics, chemistry, computer science, and mechanical engineering) and treatment-by-discipline interaction terms for each discipline studied. Interaction terms are ordered from largest to smallest treatment effect estimate. Models 1, 3, and 5 include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted) and (2) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted). Models 2, 4, and 6 include the same fixed effects but also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URM in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses.

<sup>†</sup> $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

**Table S9. Confidence Sets for Rankings of Discipline-Specific Treatment Effects on Black Speaker Representation in the 2024-25 Academic Year**

|                                                  | DV: % of Seminar Speakers Who Were Black |             | DV: Number of Seminar Speakers Who Were Black |             | DV: Did Seminar Feature at Least One Black Speaker? |             |
|--------------------------------------------------|------------------------------------------|-------------|-----------------------------------------------|-------------|-----------------------------------------------------|-------------|
|                                                  | (1)                                      | Lower Bound | (2)                                           | Lower Bound | (3)                                                 | Lower Bound |
| URM Directory Treatment × Mechanical Engineering | 3.823**                                  | <b>1</b>    | 0.574**                                       | <b>1</b>    | 0.338***                                            | <b>1</b>    |
|                                                  | (1.191)                                  |             | (0.181)                                       |             | (0.101)                                             |             |
| URM Directory Treatment × Physics                | 1.635*                                   | <b>1</b>    | 0.143*                                        | 2           | 0.093*                                              | 2           |
|                                                  | (0.649)                                  |             | (0.067)                                       |             | (0.046)                                             |             |
| URM Directory Treatment × Chemistry              | 0.453                                    | 2           | -0.052                                        | 2           | 0.063                                               | 2           |
|                                                  | (0.928)                                  |             | (0.116)                                       |             | (0.063)                                             |             |
| URM Directory Treatment × Mathematics            | 0.195                                    | 3           | 0.069                                         | 2           | 0.020                                               | 2           |
|                                                  | (0.381)                                  |             | (0.049)                                       |             | (0.028)                                             |             |
| URM Directory Treatment × Computer Science       | 0.150                                    | 2           | -0.027                                        | 2           | -0.025                                              | 2           |
|                                                  | (0.847)                                  |             | (0.088)                                       |             | (0.073)                                             |             |
| Observations                                     | 1,686                                    |             | 1,686                                         |             | 1,686                                               |             |
| Adjusted R <sup>2</sup>                          | 0.151                                    |             | 0.227                                         |             | 0.268                                               |             |

*Notes:* This table reports the results of three OLS regressions estimating discipline-specific treatment effects on Black speaker representation, along with the lower bounds of one-sided, 95% confidence sets for each discipline's rank, following the ranking inference methodology of Mogstad, Romano, Shaikh, and Wilhelm (2024). A lower bound of 1 (shown in bold) indicates that the discipline cannot be excluded from being the true top-ranked discipline at the 95% confidence level. Model 1 predicts the percentage of Black speakers in a given seminar series in the 2024-25 academic year. Model 2 predicts the number of Black speakers in a given seminar series in the 2024-25 academic year. Model 3 predicts whether a given seminar series featured at least one Black speaker in the 2024-25 academic year. All models suppress the intercept and include indicators for all five disciplines (chemistry, computer science, mathematics, mechanical engineering, and physics) and treatment-by-discipline interaction terms for each discipline. All models include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24–28, 2024; the tenth batch is omitted) and (2) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2–3 seminar series, 4–5 seminar series, 6–7 seminar series, 8–11 seminar series, 12–17 seminar series; an indicator for hosting 18–28 seminar series is omitted), and also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses. † $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

**Table S10. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation in STEM Seminars by Academic Discipline (Separate Regressions) in the 2024-25 Academic Year**

|                                                              | DV: % of AY24-25 Seminar Speakers Who Were Black |                        |                      |                       |                      | DV: Number of AY24-25 Seminar Speakers Who Were Black |                      |                     |                     |                      | DV: Did AY24-25 Seminar Have at Least One Black Speaker? |                       |                     |                      |                      |
|--------------------------------------------------------------|--------------------------------------------------|------------------------|----------------------|-----------------------|----------------------|-------------------------------------------------------|----------------------|---------------------|---------------------|----------------------|----------------------------------------------------------|-----------------------|---------------------|----------------------|----------------------|
|                                                              | <i>Math</i>                                      | <i>Phys.</i>           | <i>CS</i>            | <i>Chem.</i>          | <i>M.E.</i>          | <i>Math</i>                                           | <i>Phys.</i>         | <i>CS</i>           | <i>Chem.</i>        | <i>M.E.</i>          | <i>Math</i>                                              | <i>Phys.</i>          | <i>CS</i>           | <i>Chem.</i>         | <i>M.E.</i>          |
|                                                              | (1)                                              | (2)                    | (3)                  | (4)                   | (5)                  | (6)                                                   | (7)                  | (8)                 | (9)                 | (10)                 | (11)                                                     | (12)                  | (13)                | (14)                 | (15)                 |
| URM Directory Treatment                                      | 0.5052<br>(0.3865)                               | 1.7982**<br>(0.6662)   | -0.2852<br>(0.8233)  | 0.1176<br>(0.8979)    | 2.6193*<br>(1.1751)  | 0.1211*<br>(0.0540)                                   | 0.1571*<br>(0.0629)  | -0.0840<br>(0.0823) | -0.0044<br>(0.1059) | 0.5403*<br>(0.2230)  | 0.0364<br>(0.0263)                                       | 0.1213*<br>(0.0472)   | -0.0540<br>(0.0746) | 0.0330<br>(0.0646)   | 0.2827*<br>(0.1092)  |
| # of Faculty in Dept. of Seminar                             | -0.0355***<br>(0.0097)                           | -0.0317<br>(0.0222)    | 0.0082<br>(0.0203)   | 0.0092<br>(0.0386)    | -0.0120<br>(0.0314)  | -0.0047***<br>(0.0013)                                | -0.0016<br>(0.0028)  | 0.0011<br>(0.0018)  | 0.0084<br>(0.0064)  | -0.0059<br>(0.0048)  | -0.0028**<br>(0.0009)                                    | -0.0019<br>(0.0023)   | 0.0009<br>(0.0017)  | 0.0074*<br>(0.0033)  | -0.0021<br>(0.0029)  |
| % of Faculty in Dept. of Seminar Who Are Female              | 0.6275<br>(2.5146)                               | 6.3950<br>(5.2173)     | -3.0542<br>(4.4537)  | 10.8210†<br>(6.1762)  | 11.8534<br>(10.2317) | 0.1410<br>(0.3066)                                    | 0.0475<br>(0.5856)   | -0.4183<br>(0.6201) | 1.1469<br>(0.8799)  | 0.5617<br>(1.4417)   | 0.1901<br>(0.2043)                                       | 0.3481<br>(0.4425)    | -0.3493<br>(0.4786) | 0.7853<br>(0.4895)   | 1.2254<br>(0.8874)   |
| % of Faculty in Dept. of Seminar Who Are URM                 | 12.7952<br>(7.9187)                              | -22.5320**<br>(8.5205) | 16.4033<br>(11.8043) | 26.2511†<br>(14.1340) | 10.0459<br>(13.2532) | 1.3449<br>(1.1704)                                    | -1.4807*<br>(0.6804) | 0.0378<br>(1.4861)  | 2.7893†<br>(1.5624) | -2.1961<br>(1.9582)  | 0.1733<br>(0.4975)                                       | -1.3643**<br>(0.4923) | 0.7958<br>(1.2160)  | 2.2395**<br>(0.7839) | 0.0432<br>(1.1186)   |
| U.S. News Ranking of Dept. of Seminar                        | 0.0069<br>(0.0084)                               | -0.0249<br>(0.0204)    | -0.0081<br>(0.0470)  | 0.0317*<br>(0.0126)   | 0.0636<br>(0.0433)   | -0.0006<br>(0.0009)                                   | -0.0067*<br>(0.0031) | -0.0035<br>(0.0030) | 0.0039†<br>(0.0023) | 0.0027<br>(0.0076)   | -0.0005<br>(0.0006)                                      | -0.0042*<br>(0.0020)  | -0.0031<br>(0.0028) | 0.0019<br>(0.0013)   | 0.0049<br>(0.0031)   |
| # of Faculty Displayed in URM Directory for Dept. of Seminar | 0.0982†<br>(0.0574)                              | -0.0100<br>(0.0730)    | 0.1143<br>(0.2340)   | 0.1423<br>(0.0976)    | 0.0623<br>(0.0738)   | 0.0049<br>(0.0079)                                    | -0.0174†<br>(0.0101) | -0.0086<br>(0.0160) | 0.0280*<br>(0.0129) | 0.0076<br>(0.0126)   | 0.0080*<br>(0.0040)                                      | -0.0074<br>(0.0063)   | -0.0054<br>(0.0152) | 0.0136*<br>(0.0066)  | 0.0098<br>(0.0059)   |
| # of Peer Departments with 20+ URM Faculty                   | 0.6701<br>(0.4938)                               | 0.1722<br>(0.2020)     | 0.1490<br>(0.1746)   |                       | 1.2536†<br>(0.6855)  | 0.0261<br>(0.0408)                                    | 0.0037<br>(0.0125)   | 0.0480†<br>(0.0289) |                     | 0.1474†<br>(0.0881)  | 0.0114<br>(0.0133)                                       | 0.0082<br>(0.0103)    | 0.0277†<br>(0.0166) |                      | 0.1743**<br>(0.0605) |
| # of Dept Reps Who Received Treatment Email                  | 0.0993**<br>(0.0308)                             | -0.0691<br>(0.2042)    | 0.1293<br>(0.4460)   | 1.1100†<br>(0.6095)   | -2.0841<br>(1.3313)  | 0.0132***<br>(0.0036)                                 | 0.0056<br>(0.0253)   | -0.0143<br>(0.0336) | 0.1147†<br>(0.0669) | -0.3344†<br>(0.1807) | 0.0105***<br>(0.0026)                                    | 0.0124<br>(0.0196)    | -0.0086<br>(0.0310) | 0.0427<br>(0.0378)   | -0.2007*<br>(0.0924) |
| Observations                                                 | 823                                              | 361                    | 143                  | 277                   | 82                   | 823                                                   | 361                  | 143                 | 277                 | 82                   | 823                                                      | 361                   | 143                 | 277                  | 82                   |
| Adjusted R <sup>2</sup>                                      | 0.039                                            | 0.015                  | -0.081               | -0.012                | 0.142                | 0.026                                                 | 0.052                | 0.012               | 0.084               | 0.248                | 0.022                                                    | 0.035                 | 0.005               | 0.117                | 0.169                |

*Notes:* This table reports the results of fifteen OLS regressions estimating the effect of treatment on Black speaker representation, estimated separately for each of the five academic disciplines (mathematics, physics, computer science, chemistry, and mechanical engineering). Models 1-5 predict the percentage of Black speakers in a seminar series in

the 2024-25 Academic Year; Models 6-10 predict the number of Black speakers in a seminar series in the 2024-25 Academic Year; Models 11-15 predict whether a department's seminar series featured at least one Black speaker in the 2024-25 Academic Year. All models include fixed effects for (1) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted) and (2) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted), and also include controls for (3) the total number of faculty in the seminar's host department in summer 2024, (4) the fraction of faculty in the seminar's host department who were URMs in summer 2024, (5) the fraction of faculty in the seminar's host department who were women in summer 2024, (6) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (7) an indicator for missing department ranking, (8) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (9) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (10) the number of recipients of our experimental email about the target seminar series. Models 4, 9, and 14 (Chemistry) exclude the control for the total number of peer departments because all Chemistry seminars share the same value for this variable (40 peer departments), leaving no within-discipline variation. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

In the Directory Control condition, Black speakers comprised only 0.94% of speakers in Mechanical Engineering seminars and 1.07% in Physics seminars, compared to 3.88% in Chemistry seminars and 1.60% in Computer Science seminars. Given that we did not find any main effect of treatment on URM or Hispanic representation overall, it is not surprising that we also find no heterogeneity across disciplines for these groups (all  $p$ 's  $> .15$ ; see Extended Tables E5–E6 in the replication repository).

**Table S11. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation in STEM Seminars as a Function of Black Faculty Representation in the Seminar's Host Department in the 2024-25 Academic Year**

|                                                                                      | DV: % of AY24-25<br>Seminar Speakers Who<br>Were Black |                       | DV: Number of AY24-25<br>Seminar Speakers Who<br>Were Black |                       | DV: Did AY24-25 Seminar<br>Have at Least One Black<br>Speaker? |                       |
|--------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|-------------------------------------------------------------|-----------------------|----------------------------------------------------------------|-----------------------|
|                                                                                      | (1)                                                    | (2)                   | (3)                                                         | (4)                   | (5)                                                            | (6)                   |
| URM Directory<br>Treatment                                                           | 0.6902*<br>(0.2970)                                    | 0.7307*<br>(0.2879)   | 0.0785*<br>(0.0374)                                         | 0.0786*<br>(0.0356)   | 0.0533*<br>(0.0218)                                            | 0.0545*<br>(0.0212)   |
| Mean-Centered # of Black<br>Faculty in Dept. of Seminar                              | 0.1694<br>(0.2268)                                     | 0.1927<br>(0.2103)    | 0.0196<br>(0.0311)                                          | 0.0223<br>(0.0277)    | 0.0262<br>(0.0206)                                             | 0.0262<br>(0.0190)    |
| URM Directory Treatment ×<br>Mean-Centered # of Black<br>Faculty in Dept. of Seminar | 0.9577*<br>(0.4728)                                    | 1.0235*<br>(0.4712)   | 0.1117†<br>(0.0613)                                         | 0.1254*<br>(0.0596)   | 0.0484†<br>(0.0289)                                            | 0.0566*<br>(0.0280)   |
| # of Faculty in Dept.<br>of Seminar                                                  |                                                        | -0.0177**<br>(0.0067) |                                                             | -0.0023*<br>(0.0010)  |                                                                | -0.0013*<br>(0.0006)  |
| % of Faculty in Dept. of<br>Seminar Who Are Female                                   |                                                        | 1.9192<br>(1.8956)    |                                                             | 0.2540<br>(0.2616)    |                                                                | 0.1598<br>(0.1627)    |
| % of Hispanic Faculty in<br>Dept. of Seminar                                         |                                                        | -3.4496<br>(3.8196)   |                                                             | -1.1864*<br>(0.5034)  |                                                                | -0.5794†<br>(0.3050)  |
| % of Native American<br>Faculty in Dept. of Seminar                                  |                                                        | 0.2491<br>(11.6194)   |                                                             | -0.3342<br>(1.1270)   |                                                                | -0.0740<br>(0.6658)   |
| U.S. News Ranking of<br>Dept. of Seminar                                             |                                                        | 0.0121†<br>(0.0065)   |                                                             | 0.0016†<br>(0.0009)   |                                                                | 0.0001<br>(0.0004)    |
| # of Faculty Displayed in<br>URM Directory for Dept.<br>of Seminar                   |                                                        | 0.0571†<br>(0.0302)   |                                                             | 0.0091*<br>(0.0037)   |                                                                | 0.0055**<br>(0.0021)  |
| # of Peer Departments with<br>20+ URM Faculty                                        |                                                        | 0.2065<br>(0.1650)    |                                                             | 0.0103<br>(0.0131)    |                                                                | 0.0062<br>(0.0070)    |
| # of Dept Reps Who<br>Received Treatment Email                                       |                                                        | 0.1151***<br>(0.0342) |                                                             | 0.0182***<br>(0.0049) |                                                                | 0.0122***<br>(0.0028) |
| Observations                                                                         | 1686                                                   | 1686                  | 1686                                                        | 1686                  | 1686                                                           | 1686                  |
| Adjusted R <sup>2</sup>                                                              | 0.033                                                  | 0.040                 | 0.055                                                       | 0.072                 | 0.044                                                          | 0.056                 |

*Notes:* This table reports the results of six OLS regressions estimating the interactive effect of treatment assignment and the number of Black faculty in the seminar's host department on Black speaker representation. The number of Black faculty in a given department is mean-centered so that the treatment main effect is interpretable as the treatment effect at the average level of Black faculty representation. Models 1, 3, and 5 include fixed effects for (1) the seminar's discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted). Models 2, 4, and 6 include the same fixed effects but also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were Hispanic in summer 2024, (3) the fraction who were Native American in summer 2024, (4) the fraction of faculty in the seminar's host department who were women in summer 2024, (5) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (6) an indicator for missing department ranking, (7) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (8) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (9) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

Consistent with the lack of robust main effects reported in the main text, we found no significant heterogeneity in the effect of treatment on overall URM or Hispanic speaker representation as a function of same-group faculty representation in the seminar's host department (see Extended Tables E11–E12 in the replication repository).

**Table S12. Linear Regressions Estimating the Effect of Treatment on Black Speaker Representation by Whether the Seminar Organizer Is Black**

|                                                              | Seminars Without a Black Organizer               |                                                       |                                                          | Seminars With a Black Organizer                  |                                                       |                                                          |
|--------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------|
|                                                              | DV: % of AY24-25 Seminar Speakers Who Were Black | DV: Number of AY24-25 Seminar Speakers Who Were Black | DV: Did AY24-25 Seminar Have at Least One Black Speaker? | DV: % of AY24-25 Seminar Speakers Who Were Black | DV: Number of AY24-25 Seminar Speakers Who Were Black | DV: Did AY24-25 Seminar Have at Least One Black Speaker? |
|                                                              | (1)                                              | (2)                                                   | (3)                                                      | (4)                                              | (5)                                                   | (6)                                                      |
| URM Directory Treatment                                      | 0.7417*                                          | 0.0756 <sup>†</sup>                                   | 0.0505*                                                  | -0.5676                                          | -0.0812                                               | -0.0035                                                  |
|                                                              | (0.3166)                                         | (0.0394)                                              | (0.0236)                                                 | (1.3451)                                         | (0.1843)                                              | (0.1123)                                                 |
| # of Faculty in Dept. of Seminar                             | -0.0156*                                         | -0.0021 <sup>†</sup>                                  | -0.0011                                                  | -0.0570 <sup>†</sup>                             | -0.0041                                               | -0.0063*                                                 |
|                                                              | (0.0078)                                         | (0.0011)                                              | (0.0007)                                                 | (0.0340)                                         | (0.0044)                                              | (0.0030)                                                 |
| % of Faculty in Dept. of Seminar Who Are Female              | 1.5760                                           | 0.1782                                                | 0.1214                                                   | 18.7832                                          | 2.6439 <sup>†</sup>                                   | 1.2640                                                   |
|                                                              | (2.1416)                                         | (0.2977)                                              | (0.1784)                                                 | (11.4886)                                        | (1.5668)                                              | (0.9464)                                                 |
| % of Faculty in Dept. of Seminar Who Are URM                 | 6.2632                                           | 0.4957                                                | 0.2659                                                   | 33.2323                                          | 0.6576                                                | 3.1692*                                                  |
|                                                              | (5.0019)                                         | (0.5974)                                              | (0.3199)                                                 | (21.2787)                                        | (2.8610)                                              | (1.4716)                                                 |
| U.S. News Ranking of Dept. of Seminar                        | 0.0120 <sup>†</sup>                              | 0.0015                                                | 0.0000                                                   | 0.0176                                           | 0.0047                                                | 0.0016                                                   |
|                                                              | (0.0070)                                         | (0.0009)                                              | (0.0005)                                                 | (0.0211)                                         | (0.0032)                                              | (0.0021)                                                 |
| # of Faculty Displayed in URM Directory for Dept. of Seminar | 0.0585 <sup>†</sup>                              | 0.0086*                                               | 0.0054*                                                  | -0.0324                                          | 0.0063                                                | 0.0003                                                   |
|                                                              | (0.0319)                                         | (0.0039)                                              | (0.0022)                                                 | (0.1049)                                         | (0.0150)                                              | (0.0097)                                                 |
| # of Peer Departments with 20+ URM Faculty                   | 0.2352                                           | 0.0142                                                | 0.0090                                                   | 0.2786                                           | 0.0924                                                | 0.0100                                                   |
|                                                              | (0.1742)                                         | (0.0140)                                              | (0.0074)                                                 | (0.4599)                                         | (0.0737)                                              | (0.0299)                                                 |
| # of Dept Reps Who Received Treatment Email                  | 0.1086**                                         | 0.0157**                                              | 0.0119***                                                | -0.0461                                          | -0.0017                                               | 0.0054                                                   |
|                                                              | (0.0415)                                         | (0.0056)                                              | (0.0034)                                                 | (0.1324)                                         | (0.0157)                                              | (0.0133)                                                 |
| Observations                                                 | 1537                                             | 1537                                                  | 1537                                                     | 149                                              | 149                                                   | 149                                                      |
| Adjusted R <sup>2</sup>                                      | 0.032                                            | 0.056                                                 | 0.045                                                    | 0.028                                            | 0.175                                                 | 0.080                                                    |

*Notes:* This table reports the results of six OLS regressions estimating the effect of treatment on Black speaker representation, estimated separately for seminars with no Black organizer (Models 1-3;  $N=1,537$ ) and seminars with at least one Black organizer (Models 4-6;  $N=149$ ). All models include fixed effects for (1) the seminar's discipline (math, physics, mechanical engineering, or computer science; an indicator for chemistry is omitted), (2) the date/time when we sent an email containing our experimental manipulation to the seminar series' organizers (here we include nine fixed effects for different email batches sent between June 24-28, 2024; the tenth batch is omitted), and (3) randomization strata determined by the number of seminar series hosted in a given department in the 2023-24 academic year (1 seminar series, 2-3 seminar series, 4-5 seminar series, 6-7 seminar series, 8-11 seminar series, 12-17 seminar series; an indicator for hosting 18-28 seminar series is omitted), and also include controls for (1) the total number of faculty in the seminar's host department in summer 2024, (2) the fraction of faculty in the seminar's host department who were URM in summer 2024, (3) the fraction of faculty in the seminar's host department who were women in summer 2024, (4) the seminar's host department's U.S. News & World Report ranking in 2024 (with missing department rankings replaced by the university's overall ranking), (5) an indicator for missing department ranking, (6) the total number of URM faculty in the set of approximately 40 peer departments eligible for inclusion in our directory, (7) the total number of peer departments eligible for inclusion in our directory (see Analysis Strategy Section of paper for discussion of eligibility criteria), and (8) the number of recipients of our experimental email about the target seminar series. Robust standard errors, clustered at the seminar series' department level, are shown in parentheses. In our sample, 89% of seminars had non-Black organizers so would, in theory, benefit from bridging connections to Black speakers. Exploratory analyses examined treatment effects among seminars with Black vs. non-Black organizers across three outcomes: the percentage of speakers who were Black, the number of Black

speakers, and the probability of inviting any Black speaker. Results are massively underpowered to compare treatment effects, but if anything suggest treatment effects were directionally stronger in seminars with non-Black organizers (% Black:  $B = 0.74$ ,  $SE = 0.32$ ,  $p = 0.019$ ; # Black:  $B = 0.08$ ,  $SE = 0.04$ ,  $p = 0.055$ ; Any Black:  $B = 0.05$ ,  $SE = 0.02$ ,  $p = 0.032$ ) rather than Black organizers (% Black:  $B = -0.57$ ,  $SE = 1.35$ ,  $p = 0.674$ ; # Black:  $B = -0.08$ ,  $SE = 0.18$ ,  $p = 0.660$ ; Any Black:  $B = -0.004$ ,  $SE = 0.112$ ,  $p = 0.975$ ). For analogous analyses of URM and Hispanic speaker outcomes by organizer race, see Extended Tables E13–E14 in the replication repository.

† $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

## Section 7: Supplemental Online Experiment

We conducted a supplemental online experiment to examine whether exposure to a directory changed beliefs about the prevalence of Black STEM professors. We also measured attention to racial diversity to examine salience as an alternative explanation for the directory's effects. The design paralleled the field experiment's comparison between a directory of professors and links to department websites under shared encouragement to diversify, and added a group that received neither the encouragement nor a resource.

### Methods

**Participants.** We recruited 600 U.S.-based participants through Prolific on September 21, 2026, for a 10-minute survey paying \$2.25, with no bonus. Participants were English-speaking, used a desktop computer with audio, had at least 1,000 prior submissions and an approval rate of at least 90%, and had not participated in our earlier studies. The recorded sample contained 599 unique randomized participants (57.3% women, 41.2% men, 1.5% non-binary, age  $M = 44.22$ ,  $SD = 13.23$ ). Of these, 554 provided a valid belief response and entered the primary analysis. Our preregistered target (AsPredicted #312433) was 600 valid belief responses, and the final analysis sample fell 46 short of this target. No responses from the separate development pilot were included. The analysis files are available in our anonymous replication repository.

Participants had to pass the initial attention checks before assignment. Following the analysis plan, we excluded invalid Prolific IDs and exits before assignment and retained the first randomized response for duplicate IDs. Recorded belief responses were eligible even if participants did not finish. Of 669 recorded responses that began during this fielding, 70 ended before assignment. All 599 randomized responses had valid IDs, were unique, and were marked

finished. The *Pure Control*, *Directory Control*, and *URM Directory Treatment* conditions contained 199, 191, and 209 respondents, respectively, of whom 187, 176, and 191 provided valid beliefs. Of the 45 missing numeric beliefs, 43 were blank and two contained invalid nonnumeric or out-of-range inputs. All occurred when the belief page timed out. We did not impute these responses or recover them from comments.

**Procedure.** Participants were told to imagine that they were helping to organize a national conference on the future of science. The organizers were assembling a six-person panel of professors titled “The Future of Science” and wanted a strong panel reflecting the range of people working in science, technology, engineering, and mathematics (STEM) today. Participants were asked to nominate six professors from any STEM field and were told they could look people up online before deciding. Two comprehension questions followed, and a wrong answer led to a clarification page rather than exclusion.

Participants were then randomly assigned using a 1:1:1 target allocation to the *Pure Control*, *Directory Control*, or *URM Directory Treatment* condition. In every condition, participants were first encouraged on the nomination page to look people up online (e.g., Google, ChatGPT, or any other tool). In the *Pure Control* condition, participants received no additional resource. In the *Directory Control* condition, participants also saw the sentence “To help you ensure your panel has some racial diversity, here is a list of 20 leading STEM departments at universities across the United States, each with a link to its faculty directory,” and the link opened a table of those 20 departments. In the *URM Directory Treatment* condition, participants instead saw a sentence offering “a list of 20 professors from leading STEM departments at universities across the United States,” and the link opened a directory of 20 Black professors, one from each of the same 20 departments. The professors appeared in a fixed order with their

photograph, name, university, department, title, and area of expertise. Neither the sentence nor the directory described the professors as Black.

Both directory conditions therefore received the same encouragement to ensure their panel had some racial diversity. These conditions parallel the field experiment's two conditions, with the online URM directory limited to Black professors. We added *Pure Control* to assess whether providing a search resource alongside encouragement to diversify changed beliefs or salience relative to a baseline without either component. Comparing the two directory conditions then tested whether the directory changed beliefs or salience beyond *Directory Control*. Resource use was optional, and nominations were not restricted to the listed departments or professors. Link tracking recorded openings for 96 of 191 *Directory Control* participants (50.3%) and 121 of 209 *URM Directory Treatment* participants (57.9%). These counts may not capture all openings. All analyses retained participants in their assigned conditions regardless of resource use.

**Measures.** After making their nominations, participants were told not to use a search engine or artificial intelligence tool to answer the next question and that the page would advance automatically after 15 seconds. Our primary dependent measure elicited posterior beliefs by asking, “Of all the professors in STEM departments at colleges and universities in the United States, what is your best estimate of the percentage who are Black or African American?” Participants could enter a value from 0 to 100, and a response was not required. We elicited beliefs only after nominations to avoid drawing additional attention to racial representation before the task. This choice also limits demand or consistency concerns associated with repeated belief elicitation (Haaland et al. 2023). Our analyses therefore compare post-treatment beliefs across conditions rather than within-person changes from an elicited baseline.

We assessed salience because drawing attention to diversity can increase the selection of women in hiring decisions (Chang et al. 2020). After the belief question, participants responded to four items in random order on a seven-point scale from 1 (Strongly disagree) to 7 (Strongly agree). Following the stem “While I was deciding who to nominate...”, three items stated, “My mind was focused on the racial diversity of the panel,” “I paid a lot of attention to the racial diversity of the panel,” and “I focused a great deal of attention on the racial diversity of the panel.” These items were adapted from Rich et al. (2010) and averaged when all three were observed (Cronbach's  $\alpha = .975$  in the analysis sample). The separate item, adapted from Chang et al. (2020), stated, “I considered how my nominations would influence the racial diversity of the panel when making my decisions.” Responses were required, and all 599 randomized participants answered all four items. Participants then classified the race of each professor they nominated and reported their demographics.

**Analysis strategy.** Following our preregistration, we ran ordinary least squares regressions (OLS) with indicators for assignment to *URM Directory Treatment* and *Directory Control* as our predictors and *Pure Control* as the omitted reference group. We compared the two coefficients using a Wald test. Our primary outcome was the belief estimate, and we analyzed the three-item salience mean and the separate consideration item as secondary outcomes. The preregistered exclusion of missing beliefs yielded the same 554-person sample for all three outcomes. We also analyzed both salience measures among all 599 randomized participants as sensitivity analyses. We used ordinary model-based standard errors, two-sided tests, and *t*-based 95% confidence intervals. We did not adjust *p*-values for multiple comparisons or exclude participants on the basis of their nominations or resource use.

## Results

**Belief estimates.** As shown in Table S13, participants estimated a significantly higher percentage of Black STEM professors in the *URM Directory Treatment* condition ( $M = 26.58$ ,  $SD = 23.13$ ) than in *Directory Control* ( $M = 14.74$ ,  $SD = 12.73$ ,  $B = 11.83$  percentage points,  $SE = 1.74$ , 95% CI [8.42, 15.24], Wald  $F(1, 551) = 46.39$ ,  $p < .001$ ) or *Pure Control* ( $M = 13.47$ ,  $SD = 10.95$ ,  $B = 13.11$ ,  $SE = 1.71$ , 95% CI [9.75, 16.46],  $p < .001$ ). We found no significant difference between the *Directory Control* and *Pure Control* conditions ( $B = 1.27$ ,  $SE = 1.75$ , 95% CI [-2.16, 4.70],  $p = .466$ ). Beliefs were measured on a 0–100 percentage scale, and these contrasts are expressed in percentage points.

**Salience.** As shown in Table S14, Models 1–2, both directory conditions produced higher salience than *Pure Control* on both measures (all  $ps < .001$ ). For the three-item measure, the means were 4.37 ( $SD = 1.90$ ) in *URM Directory Treatment*, 4.68 ( $SD = 1.86$ ) in *Directory Control*, and 2.68 ( $SD = 1.79$ ) in *Pure Control*. We found no significant difference between *URM Directory Treatment* and *Directory Control* ( $B = -0.32$ ,  $SE = 0.19$ , 95% CI [-0.70, 0.06], Wald  $F(1, 551) = 2.69$ ,  $p = .101$ ). For the separate consideration item, the corresponding means were 4.44 ( $SD = 2.02$ ), 4.81 ( $SD = 1.95$ ), and 2.87 ( $SD = 1.85$ ). Again, we found no significant difference between the two directory conditions ( $B = -0.37$ ,  $SE = 0.20$ , 95% CI [-0.77, 0.03], Wald  $F(1, 551) = 3.28$ ,  $p = .071$ ).

**Sensitivity analyses.** The pattern was unchanged when we included all 599 randomized respondents with observed salience (Table S14, Models 3–4). Both directory conditions exceeded *Pure Control* on both measures (all  $ps < .001$ ). We found no significant difference between *URM Directory Treatment* and *Directory Control* for either the three-item measure ( $B = -0.23$ ,  $SE = 0.19$ , 95% CI [-0.60, 0.13], Wald  $F(1, 596) = 1.57$ ,  $p = .211$ ) or the separate

consideration item ( $B = -0.29$ ,  $SE = 0.20$ , 95% CI  $[-0.67, 0.10]$ , Wald  $F(1, 596) = 2.14$ ,  $p = .144$ ). This analysis relaxes the valid-belief restriction used for the preregistered secondary analyses.

## Discussion

The directory increased beliefs about the prevalence of Black STEM professors relative to both comparison groups. Both directory conditions also increased salience relative to *Pure Control*, suggesting that providing a search resource alongside encouragement to diversify can draw attention to diversity. However, we found no significant difference in salience between the *URM Directory Treatment* and *Directory Control* conditions. Since we encouraged organizers in both conditions of our field experiment to diversify their seminars, these findings suggest that drawing attention to diversity may have contributed to their efforts in both conditions. We interpret the belief and salience results as more consistent with belief updating than with an additional increase in diversity salience as an explanation for the indirect effect we observed in the field: increased selection of Black speakers who were not listed in the directory. The supplemental online study provides complementary evidence, rather than a direct test of mediation of the field effect. The comparisons do not establish salience equivalence or isolate the effect of encouragement alone. Our online measure captures beliefs about the overall prevalence of Black STEM professors rather than the number meeting seminar organizers' qualification standards, and higher estimates need not be more accurate.

## Section 8: References

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**Table S13. Linear Regression Estimating the Effect of the URM Directory on Beliefs About the Prevalence of Black STEM Professors**

| Dependent variable                                        | Belief estimate (0–100%)<br>(1) |
|-----------------------------------------------------------|---------------------------------|
| <i>URM Directory Treatment</i>                            | 13.11***<br>(1.71)              |
| <i>Directory Control</i>                                  | 1.27<br>(1.75)                  |
| Constant                                                  | 13.47***<br>(1.22)              |
| Observations                                              | 554                             |
| R-squared                                                 | 0.114                           |
| Adjusted R-squared                                        | 0.111                           |
| <b><i>URM Directory Treatment – Directory Control</i></b> |                                 |
| Difference                                                | 11.83                           |
| Standard error                                            | 1.74                            |
| 95% confidence interval                                   | [8.42, 15.24]                   |
| Wald <i>F</i>                                             | 46.39                           |
| Wald <i>p</i>                                             | < .001                          |

**Notes:** This table reports an OLS regression predicting participants' estimates of the percentage of U.S. STEM professors who are Black in the supplemental online experiment. The sample contains 554 participants with valid belief responses. Coefficients are percentage points on a 0–100 scale. The predictors are indicators for assignment to *URM Directory Treatment* and *Directory Control*, with *Pure Control* as the omitted condition. All models include an intercept. Ordinary model-based standard errors appear in parentheses. The lower panel compares the two condition coefficients using a one-restriction Wald *F* test. Tests are two-sided, confidence intervals use the *t* distribution, and no multiplicity adjustment is applied.

†*p* < .10, \**p* < .05, \*\**p* < .01, \*\*\**p* < .001.

**Table S14. Linear Regressions Estimating the Effect of the URM Directory on Racial-Diversity Salience**

| Dependent variable                                            | Main secondary analyses<br>Valid-belief sample |                                   | Sensitivity analyses<br>All randomized respondents |                                   |
|---------------------------------------------------------------|------------------------------------------------|-----------------------------------|----------------------------------------------------|-----------------------------------|
|                                                               | Three-item<br>attention mean                   | Separate<br>consideration<br>item | Three-item<br>attention mean                       | Separate<br>consideration<br>item |
| Model                                                         | (1)                                            | (2)                               | (3)                                                | (4)                               |
| <i>URM Directory Treatment</i>                                | 1.68***<br>(0.19)                              | 1.57***<br>(0.20)                 | 1.70***<br>(0.18)                                  | 1.59***<br>(0.19)                 |
| <i>Directory Control</i>                                      | 2.00***<br>(0.19)                              | 1.94***<br>(0.20)                 | 1.94***<br>(0.19)                                  | 1.87***<br>(0.20)                 |
| Constant                                                      | 2.68***<br>(0.14)                              | 2.87***<br>(0.14)                 | 2.71***<br>(0.13)                                  | 2.88***<br>(0.14)                 |
| Observations                                                  | 554                                            | 554                               | 599                                                | 599                               |
| R-squared                                                     | 0.184                                          | 0.159                             | 0.177                                              | 0.151                             |
| Adjusted R-squared                                            | 0.181                                          | 0.156                             | 0.175                                              | 0.149                             |
| <b><i>URM Directory Treatment –<br/>Directory Control</i></b> |                                                |                                   |                                                    |                                   |
| Difference                                                    | –0.32                                          | –0.37                             | –0.23                                              | –0.29                             |
| Standard error                                                | 0.19                                           | 0.20                              | 0.19                                               | 0.20                              |
| 95% confidence interval                                       | [–0.70, 0.06]                                  | [–0.77, 0.03]                     | [–0.60, 0.13]                                      | [–0.67, 0.10]                     |
| Wald <i>F</i>                                                 | 2.69                                           | 3.28                              | 1.57                                               | 2.14                              |
| Wald <i>p</i>                                                 | .101                                           | .071                              | .211                                               | .144                              |

**Notes:** This table reports four OLS regressions predicting racial-diversity salience in the supplemental online experiment. Models 1 and 3 use the mean of three attention items adapted from Rich et al. (2010). Models 2 and 4 use a separate consideration item adapted from Chang et al. (2020). Both outcomes range from 1 to 7. Models 1–2 use the preregistered valid-belief sample ( $N = 554$ ). Models 3–4 are sensitivity analyses using all recorded randomized respondents with observed salience ( $N = 599$ ). Coefficients are scale points. The predictors are indicators for assignment to *URM Directory Treatment* and *Directory Control*, with *Pure Control* as the omitted condition. All models include an intercept. Ordinary model-based standard errors appear in parentheses. The lower panel compares the two condition coefficients using a one-restriction Wald *F* test. Tests are two-sided, confidence intervals use the *t* distribution, and no multiplicity adjustment is applied.

† $p < .10$ , \* $p < .05$ , \*\* $p < .01$ , \*\*\* $p < .001$ .